

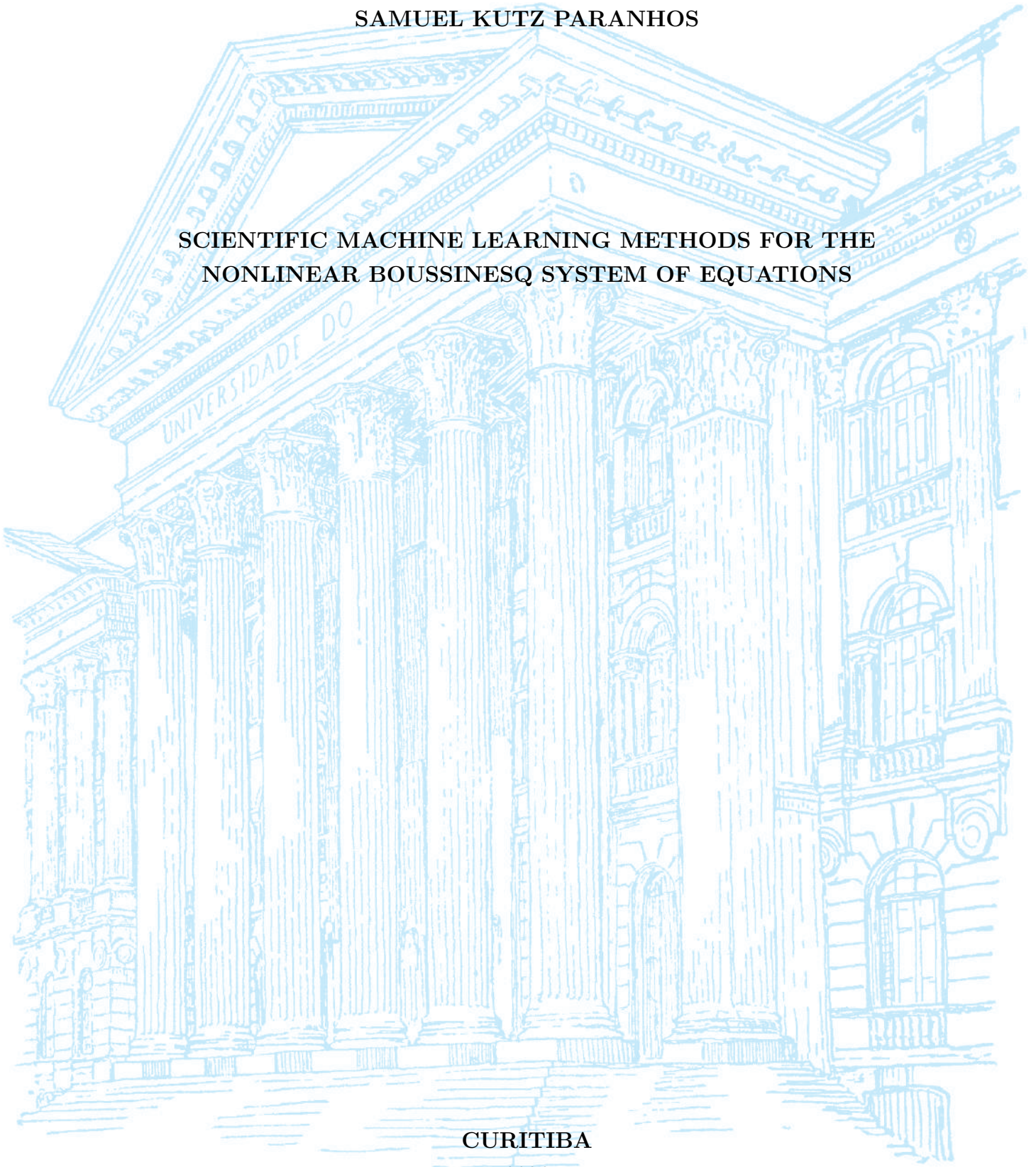
UNIVERSIDADE FEDERAL DO PARANÁ

SAMUEL KUTZ PARANHOS

SCIENTIFIC MACHINE LEARNING METHODS FOR THE
NONLINEAR BOUSSINESQ SYSTEM OF EQUATIONS

CURITIBA

2026



SAMUEL KUTZ PARANHOS

TRABALHO DE CONCLUSÃO DE CURSO

**SCIENTIFIC MACHINE LEARNING METHODS FOR THE
NONLINEAR BOUSSINESQ SYSTEM OF EQUATIONS**

Trabalho apresentado como requisito parcial à obtenção do grau de Bacharel em
Matemática Industrial no Departamento de Matemática da Universidade Federal
do Paraná.

Área de concentração: Matemática Industrial.

Orientador: Prof. Roberto Ribeiro.

Título do Projeto: Scientific Machine Learning Methods for the Nonlinear
Boussinesq System of Equations.

CURITIBA

2026

RESUMO

Este trabalho investiga a eficácia de métodos de Scientific Machine Learning (SciML) na resolução e aproximação do sistema não linear de Boussinesq, um sistema de Equações Diferenciais Parciais (EDPs) utilizado para modelar a dinâmica de ondas aquáticas. Neste estudo, o sistema de Boussinesq será tratado como uma EDP dependente de dois parâmetros: os parâmetros de não linearidade e de dispersão. O objetivo é analisar o uso do ecossistema contemporâneo de ferramentas desenvolvido em torno do recente crescimento de Machine Learning como complemento aos métodos numéricos clássicos empregados na solução do sistema de Boussinesq. Foi utilizado o método pseudoespectral para gerar um conjunto de dados consistindo de soluções de referência do sistema de Boussinesq para diversos parâmetros e é então empregado como base de treinamento para três arquiteturas de SciML investigadas neste trabalho: Redes Neurais Informadas pela Física (PINNs), treinadas a um único parâmetro tanto com restrições de dados quanto de forma puramente física; Operadores Neurais de Fourier (FNO), operando em regime puramente supervisionado a vários parâmetros com uso exclusivo de dados sem nenhuma informação do operador diferencial da EDP; e Operadores Neurais Informados pela Física (PINO), que consiste em uma FNO com a inclusão do operador diferencial da EDP na função de perda da arquitetura. Como métricas de eficiência dessas arquiteturas, utilizamos a evolução do erro espectral e do erro relativo ao longo do espaço-tempo com relação a solução de referência. Como resultados, percebe-se que as PINNs tradicionais, especialmente quando treinadas sem o auxílio de dados, sofrem de um fenômeno conhecido como viés espectral, apresentando tendência a aprender primeiramente as componentes de baixa frequência da solução e grandes dificuldades em capturar detalhes essenciais de alta frequência à medida que a não linearidade aumenta. Em contrapartida, por aprenderem diretamente no espaço das frequências, os modelos de aprendizado de operadores, como o FNO e o PINO treinado com dados, atenuam substancialmente essa limitação nas baixas e médias frequências, embora nenhum método a elimine por completo: um erro residual de alta frequência persiste em todas as arquiteturas, e o viés reaparece de forma mais acentuada quando o PINO é treinado de maneira puramente física. Observa-se ainda que a inclusão do resíduo físico confere estabilidade temporal, suprimindo o artefato de Gibbs que degrada o FNO no instante final. Assim, concluímos que a inclusão de dados na função de perda constitui a estratégia mais eficaz para mitigar — ainda que não anular — o viés espectral comumente presente nessas arquiteturas.

Palavras-chave: Scientific Machine Learning; Equações Diferenciais Parciais; Sistema de Boussinesq; Redes Neurais Informadas pela Física.

ABSTRACT

This work investigates the effectiveness of Scientific Machine Learning (SciML) methods in solving and approximating the nonlinear Boussinesq system, a system of Partial Differential Equations (PDEs) used to model water wave dynamics. In this study, the Boussinesq system will be treated as a PDE dependent on two parameters: the nonlinearity and dispersion parameters. The objective is to analyze the use of the contemporary ecosystem of tools developed around the recent growth of Machine Learning as a complement to the classical numerical methods employed in solving the Boussinesq system. The pseudospectral method was used to generate a dataset consisting of reference solutions of the Boussinesq system for various parameters and is then employed as the training basis for three SciML architectures investigated in this work: Physics-Informed Neural Networks (PINNs), trained at a single parameter both with data constraints and purely physics-informed; Fourier Neural Operators (FNOs), operating in a purely supervised regime at multiple parameters using only data without any differential operator information from the PDE; and Physics-Informed Neural Operators (PINOs), which consist of an FNO with the inclusion of the PDE differential operator in the architecture loss function. As efficiency metrics for these architectures, we use the evolution of spectral error and relative error over space-time with respect to the reference solution. As results, it is observed that traditional PINNs, especially when trained without the aid of data, suffer from a phenomenon known as spectral bias, showing a tendency to learn the low-frequency components of the solution first and great difficulty capturing essential high-frequency details as nonlinearity increases. In contrast, by learning directly in the frequency space, operator learning models such as FNO and PINO trained with data attenuate this limitation substantially in the low and mid frequencies, although no method removes it entirely: a residual high-frequency error persists across all architectures, and the bias re-emerges more strongly when PINO is trained in a purely physics-informed manner. We further observe that the physics residual confers temporal stability, suppressing the Gibbs artifact that degrades the FNO at the final time. Thus, we conclude that including data in the loss function is the most effective strategy to mitigate — though not to eliminate — the spectral bias commonly present in these architectures.

Keywords: Scientific Machine Learning; Partial Differential Equations; Boussinesq System; Physics-Informed Neural Networks.

ACKNOWLEDGEMENTS

I would like to thank CNPq, the LabFluid laboratory, and the Federal University of Paraná (UFPR) for their support, infrastructure, and guidance during this research.

This work was supported by CNPq and developed with the collaboration of the LabFluid research group at UFPR.



CONTENTS

RESUMO	3
ABSTRACT	4
ACKNOWLEDGEMENTS	5
LIST OF FIGURES	7
LIST OF ABBREVIATIONS	10
LIST OF SYMBOLS	11
1 INTRODUCTION	13
1.1 Literature Overview on Scientific Machine Learning for PDEs	13
1.2 The Boussinesq System of Equations	16
1.3 Research Goals	18
1.4 Work Structure	19
2 MATHEMATICAL BACKGROUND	20
2.1 The Boussinesq System	20
2.1.1 Recovering the Wave Equation	21
2.2 Pseudospectral Method	22
2.2.1 Spatial Discretization and the Discrete Fourier Transform	23
2.2.2 Spectral Formulation of the Boussinesq System	23
2.2.3 Time Evolution via Runge-Kutta 4	25
2.2.4 Training Dataset	25
2.3 Neural Networks	26
2.3.1 Multilayer Perceptrons	26
2.3.2 Gradient Flow	27
2.3.3 Neural Tangent Kernel	28
2.3.4 Spectral Bias	30
2.3.5 Physics-Informed Neural Networks	35
2.3.6 Spectral Bias in Physics-Informed Neural Networks	36

2.4	Neural Operators	37
2.4.1	General Definition	37
2.4.2	Fourier Neural Operators	38
2.4.3	Physics-Informed Neural Operators	40
3	METHODOLOGY	42
3.1	Setup, Solver, and Reproducibility	42
3.2	Building the Parametric Dataset	43
3.3	A Common Yardstick	45
3.4	A Ladder of Approximators	47
3.4.1	The Plain Regressor: MLP	47
3.4.2	Adding the Equation: PINN	48
3.4.3	Learning the Operator: FNO	48
3.4.4	Physics on the Operator: PINO	49
3.5	Probing the Spectral Bias	50
3.5.1	The Kernel of a Real PINN	50
3.5.2	A Toy Model Where the Theory Is Exact	51
3.6	Configuration Summary	52
4	NUMERICAL RESULTS	53
4.1	The Price of Each Approach	53
4.2	Spectral Bias, From Theory to the Network	54
4.3	Learning by Frequency Band	57
4.4	Reproducing the Spectrum	59
4.5	Where Physics Earns Its Keep	66
4.6	Two Stress Tests	69
4.7	What the Experiments Show	71
5	CONCLUSION	73
5.1	A Verdict on Each Approach	73
5.2	What This Study Cannot Claim	75
5.3	Directions Forward	76

LIST OF FIGURES

2.1	Initial condition $\eta(x, 0) = A \operatorname{sech}^2(x)$ for $A = 1$	21
2.2	Example of MLP architecture of input (x, t) , with $L_{\text{MLP}} = 3$ hidden layers and $N_{\text{MLP}} = 6$ neurons per layer.	27

4.1	Spectral-bias theory validated on the Fourier-feature model of Section 3.5.2, trained by full-batch gradient descent at $\mu = 1/\lambda_{\max}$ so that the linearized update (2.61) is exact. Measured curves (markers) overlay the theoretical predictions (lines) until the float64 floor.	55
4.2	Neural Tangent Kernel diagnostic for the Boussinesq PINN residual at $\alpha = \beta = 3.21$, for widths 8, 128 and 512.	56
4.3	Relative spectral error by frequency band during training. In every panel the low band (blue) is learned first and driven lowest, the mid band (purple) trails it, and the high band (red) is learned last and least, the signature of spectral bias. Note the mixed reference samples: the pointwise MLP and PINN are tracked at $\alpha = \beta = 3.21$, the operators at $\alpha = \beta = 0.1$	58
4.4	MLP (data only) spatial spectrum at $\alpha = \beta = 3.21$. Fitting the reference with no equation and no physics term, the network reproduces the low-wavenumber crests but undershoots the tail: the relative error climbs with wavenumber, the purest picture of spectral bias among the six models. . .	60
4.5	PINN (no data) spatial spectrum at $\alpha = \beta = 3.21$. The network cannot represent the sech^2 tail even at $t = 0$, and the relative error rises monotonically with wavenumber; the mean spectral error grows from 0.17 at $t = 0$ to 0.70 at $t = 30$	61
4.6	PINN (with data) spatial spectrum at $\alpha = \beta = 3.21$. The low-wavenumber crests are tracked more tightly than in Figure 4.5, but the high-wavenumber undershoot and the growth of the mean error in time ($0.17 \rightarrow 0.58 \rightarrow 0.68$) remain.	62
4.7	FNO spatial spectrum at $\alpha = \beta = 3.21$. Low and mid wavenumbers are tracked almost exactly (mean error 0.013 at $t = 0$), but the error migrates to the high modes and grows steadily in time, reaching a mean of 0.40 at $t = 30$	63
4.8	PINO (with data) spatial spectrum at $\alpha = \beta = 3.21$. The prediction tracks the reference across the band and, unlike the FNO, the mean error stays bounded in time ($0.077 \rightarrow 0.13 \rightarrow 0.20$): the physics residual steadies the temporal evolution.	64
4.9	PINO (no data) spatial spectrum at $\alpha = \beta = 3.21$. Without the supervised term the low- and mid-band undershoot returns (mean error $0.11 \rightarrow 0.39 \rightarrow 0.33$), confirming that the physics residual alone does not overcome spectral bias.	65

- 4.10 FNO across the nonlinearity axis ($\alpha = \beta \in \{0.1, 3.21, 4.2\}$), evaluated at resolution 256. The predicted surfaces (left) capture the two counter-propagating pulses, but the time-resolved relative error (right) spikes sharply at the final time $t = 30$: the temporal Gibbs artifact of treating time as a periodic axis. 67
- 4.11 PINO (with data) across the same nonlinearity axis and resolution. The predicted surfaces are visually indistinguishable from the FNO, but the terminal-time spike of Figure 4.10 is gone: the error rises only gently toward $t = 30$. The physics residual and the initial-condition term act as a temporal regularizer. 68
- 4.12 PINO (with data) queried zero-shot at resolutions 128, 256 and 512 at $\alpha = \beta = 3.21$. The operator returns a coherent field at every resolution, but the relative error against the pseudospectral reference *grows* with resolution (median rising from about 0.07 at 128 to about 0.16 at 512): invariance of form is not invariance of accuracy. 70

LIST OF ABBREVIATIONS

SciML	Scientific Machine Learning
ML	Machine Learning
PDE	Partial Differential Equation
PINN	Physics-Informed Neural Network
FNO	Fourier Neural Operator
PINO	Physics-Informed Neural Operator
MLP	Multilayer Perceptron
RK4	Runge–Kutta 4th-order method
DFT	Discrete Fourier Transform
FFT	Fast Fourier Transform
NTK	Neural Tangent Kernel
MSE	Mean Squared Error
SGD	Stochastic Gradient Descent
AD	Automatic Differentiation

LIST OF SYMBOLS

Domain and general operators

x	spatial coordinate
t	temporal coordinate
$u(x, t)$	PDE solution (generic)
$\mathcal{D}$	differential operator of the PDE
$\mathcal{L}$	linear differential operator; also a loss functional
∇	gradient (nabla) operator
N_x	number of spatial grid points
N_t	number of time steps
$\Delta x, \Delta t$	spatial and temporal grid spacings

Boussinesq system and spectral quantities

$\eta(x, t)$	surface elevation field
$u(x, t)$	depth-averaged horizontal velocity field
α	nonlinearity parameter of the Boussinesq system
β	dispersion parameter of the Boussinesq system
A	initial wave amplitude
L	domain half-length, $\Omega = [-L, L]$
k	wavenumber (integer mode index)
ξ_k	wavenumber $\pi k/L$ on the domain $[-L, L]$
$\omega(k)$	angular frequency / dispersion relation
$\hat{f}(k)$	Fourier coefficient of f at mode k
$\hat{f}(k, t)$	time-evolving Fourier mode of f at wavenumber k

Neural networks and Neural Tangent Kernel

θ	trainable parameter vector of a neural network
u_θ	neural network approximation of the PDE solution
$h^{(l)}$	hidden-layer activation vector at layer l
σ	activation function
$W^{(l)}, b^{(l)}$	weight matrix and bias vector at layer l
N_{MLP}	number of neurons in a hidden layer
L_{MLP}	number of layers in the neural network
$K(\theta)$	empirical Neural Tangent Kernel, $K = JJ^\top$
λ_k	NTK eigenvalue associated with mode k
$\hat{e}(k, t)$	Fourier coefficient of the training error at mode k

Neural operators

$G^\dagger, G_\theta$	target solution operator and its neural approximation
$\mathcal{K}_\ell$	integral operator of the ℓ -th FNO spectral layer
κ_ℓ	kernel function of the FNO integral operator
P	lifting operator of the FNO
Q	projection operator of the FNO
d_c	channel (width) dimension of the FNO
$R_{\ell,k}$	FNO spectral weight (discretized kernel) at layer ℓ , mode k
k_{max}	FNO mode-truncation threshold

1 INTRODUCTION

Since the start of the scientific thinking, mathematics has been one of the key languages in which we can understand the world, and Partial Differential Equations (PDEs) are one of the main dialects of the mathematical language we use to model and to simplify nature’s complexity.

When modelling these equations, we cannot always afford them to have simple closed-form solutions (that is, when they have solution at all), this exposes the need of obtaining approximated solutions coming from numerical methods for these PDEs, also giving the option to circumvent analytical complexity. This allows the extraction of predictions, comparisons with data, build simulations and so on. These classical numerical methods, widely studied for all types of differential equations, usually rely on discretizing the PDE’s domain into grids and performing temporal steps.

With the recent ascension of Machine Learning (ML) methods and a whole ecosystem of tools built around it, a natural path is to bridge these two worlds and let ML augment the numerical solution of PDEs, taking advantage of the many software frameworks developed. This is one facet of Scientific Machine Learning (SciML), the broader effort to fuse physics-based, mechanistic models with data-driven methods. Within SciML, this work focuses on the numerical solution of PDEs and, given the vast literature around the topic, restricts its review of recent works to hydrodynamics equations.

1.1 Literature Overview on Scientific Machine Learning for PDEs

The neural network building blocks underlying all of these approaches trace a line beginning with the formal neuron of McCulloch et al. (1943) and Rosenblatt’s perceptron (ROSENBLATT, 1958), the first trainable single-layer model. The key algorithmic advance that unlocked multi-layer training was the backpropagation algorithm (RUMELHART et al., 1986), making gradient-based optimization practical for deep architectures. Shortly after, the universal approximation theorem (CYBENKO, 1989; HORNIK et al., 1989) provided the theoretical guarantee for a multilayer perceptron (MLP) with a single hidden layer of sufficient width to be able to approximate any continuous function on a compact domain to arbitrary precision. This combination of practical and universal

approximation established the MLP as the backbone of modern deep learning. The term neural network itself reflects the biological metaphor at the heart of these models, in which McCulloch et al. (1943) drew an explicit analogy to the firing of biological neurons, and this vocabulary propagated through the field, causing MLPs to be routinely and interchangeably referred to as artificial neural networks (ANNs) or feedforward neural networks. The idea of “deep” in deep learning, popularized following the breakthrough results of LeCun et al. (2015), simply denotes an MLP with many hidden layers, a regime in which representational capacity grows substantially and where the backpropagation-based training paradigm proves to be exceptionally efficient.

These ML foundations become the basis of a scientific methodology once the methods are coupled to the physical laws that govern a problem of interest. This is the premise of Scientific Machine Learning, a term consolidated by Baker et al. (2019) to name the discipline at the intersection of scientific computing and machine learning, and surveyed in breadth by Karniadakis et al. (2021). Its unifying goal is to combine the interpretability, generalization, and inductive biases of physics-based models with the flexibility and data-adaptivity of ML, rather than treating the network as a pure black box. The methods span several families. Some recover governing equations directly from data, as in the Sparse Identification of Nonlinear Dynamics (SINDy) (BRUNTON et al., 2016). Others embed differential-equation structure into the network itself, as in Neural Ordinary Differential Equations (neural ODEs) (CHEN, R. T. Q. et al., 2018) and Universal Differential Equations (UDEs), in which a trainable network stands in for the unknown terms of an otherwise mechanistic model (RACKAUCKAS et al., 2020). These tools have been carried across a wide range of domains, from global weather and climate forecasting (BONEV et al., 2023) to molecular dynamics, materials design, and turbulence modelling (KARNIADAKIS et al., 2021). Among all these directions, the one that concerns us here is the use of SciML to obtain numerical solutions of PDEs and, more specifically, of the hydrodynamics equations that govern water waves. Given the vastness of the literature, the remainder of this review is devoted to that subarea.

The contemporary landscape of SciML for PDEs was reshaped by one idea in particular, the Physics-Informed Neural Networks (PINNs) (RAISSI et al., 2019), that is, the embedding of the governing PDE residual directly into the training loss of a Neural Network, working as regularization term for not only fitting data, but also respecting the underlying physics. In hydrodynamics, a growing body of work has applied PINNs to water flow (DE PINA et al., 2021, 2023), shallow-water dynamics on the sphere (BIHLO et al., 2022), and inverse problems in strongly nonlinear wave systems (JAGTAP et al., 2022). The consistent result that we can address from reviews (CUOMO et al., 2023) is that PINNs are most effective when observational data are sparse and the task is

parameter identification, namely inverse problems, which are known for usually being not well-posed problems. But, for a reliable precise forward simulation at scale, they do not yet rival classical solvers. In our vision, these approaches serve best to complement them.

As a natural extension from neural networks, there is a recent development in learning maps between infinite-dimensional function spaces, in which the theoretical foundation was popularized by T. Chen et al. (1995), who extended the universal approximation theorem to nonlinear operator, in other words, a shallow network can approximate any continuous operator between Banach spaces to arbitrary precision. Bringing this idea into practical terms, Lu et al. (2021) introduced DeepONet, the first deep architecture expressly designed for learning operators coming from differential equations, demonstrating it across a range of ODEs and PDEs. Building on this line of work, operator learning puts aside the single-instance limitation of PINNs by training a model to approximate the full mapping from parameters and initial data to the solution, so that at inference time a new solution costs only a forward pass for the network. Parallel to DeepONet, Fourier Neural Operator (FNO) (LI et al., 2021) is one of the most widely adopted instance of this idea, it parametrizes an integral kernel in Fourier space, achieving speed-ups over other neural operators and is discretization-independent, the so-called zero-shot super-resolution. Subsequent works explored alternative bases like wavelets (GUPTA et al., 2021), but FNO was validated on many types of equations such as dispersive and soliton wave equations (PEI et al., 2022; ZHONG et al., 2023), was also caled to geophysical problems spanning regional ocean modeling, shallow-water emulation, and global weather forecasting (BONEV et al., 2023; LKHAGVASUREN et al., 2024; UCHIDA et al., 2025; PATEL et al., 2025; YU et al., 2025).

Following the idea for PINNs, The Physics-Informed Neural Operator (PINO) (LI et al., 2023a) combines operator learning with PDE-residual constraints, cutting the labelled-data requirement while preserving physical consistency. Applications to shallow-water systems (ROSOFISKY et al., 2023) confirmed that this hybrid regime inherits the generalization of FNO without its data hunger. Broad surveys of the field (TODO, 2023) identify these physics-data approaches working together as the most promising SciML direction, although highlighting turbulence modeling, scalability, and noisy-data robustness as open challenges.

A known limitation that cuts across all SciML architectures is the so-called spectral bias, the tendency of gradient-based optimizers to learn the low-frequency components of a solution faster than the high-frequency ones (RAHAMAN et al., 2019; XU et al., 2020). The theoretical account of this behavior came with the Neural Tangent Kernel (NTK) (JACOT et al., 2018), which showed that a sufficiently wide network trains, to leading order, as a linear model governed by a kernel that stays essentially fixed dur-

ing optimization, the so-called lazy-training regime. In this picture each mode of the solution is learned at a rate set by its associated NTK eigenvalue, so the wide disparity between the largest and smallest eigenvalues translates directly into the frequency imbalance seen during training (ARORA et al., 2019; CAO et al., 2021; BORDELON et al., 2020). Wang, Yu, et al. (2022) carried this analysis to PINNs, showing that the eigenvalue disparity between the PDE-residual and initial-condition loss components is what drives their spectral bias. This pathology is especially consequential for dispersive wave problems, where the solution energy is broadly distributed across wavenumbers. A recent systematic study (KHODAKARAMI et al., 2026) showed that second-order optimizers combined with spectrum-aware loss formulations substantially mitigate it.

As far as this review reveals, SciML methods have been applied to a broad class of shallow-water and dispersive wave equations. To the best of our knowledge, however, no study has systematically evaluated FNO, PINO, or PINNs on the parametric 1D Boussinesq system, where both the nonlinearity coefficient α and the dispersion coefficient β vary simultaneously. This gap motivates our effort to implement and evaluate these models with the goal of exposing their details for the user aiming to bring together this new set of tools.

1.2 The Boussinesq System of Equations

For this work, we choose to implement SciML methods for a specific PDE originating from J. Boussinesq’s 1872 effort to give a theoretical account of the solitary wave of translation observed by J. Scott Russell in a canal in 1834 (BOUSSINESQ, 1872). Russell had noticed that a hump of water, set in motion by a stopping barge, could travel for miles without changing shape, a phenomenon the linear wave theory of the time could not explain. Boussinesq showed that the stability of such a wave is the result of a precise balance between nonlinear steepening and frequency dispersion, two effects that are separately destabilizing but mutually compensating in the shallow-water, long-wave regime. The mathematical structure of this balance, and of dispersive wave theory in general, are treated in depth by Whitham (1974) and Debnath (2012).

The modern two-field form of the system, for the free-surface elevation $\eta(x, t)$ and the depth-averaged horizontal velocity $u(x, t)$, was systematically derived and classified by Bona et al. (2002), who identified a four-parameter family of equivalent Boussinesq systems obtainable from the incompressible, irrotational Euler equations by asymptotic simplifications. Earlier, Peregrine (1967) had already written down the standard form of these equations for water of variable depth, establishing the notation and scaling that remain in widespread use. What makes the Boussinesq system particularly suitable as a parametric testbed for this work is that changing the nonlinearity and dispersion co-

efficients continuously deforms the solution from the near-linear, wave-packet regime to the strongly nonlinear soliton regime, giving rise to a more interesting range of behaviors within a single mathematical framework.

The specific one-dimensional form of the system used in this work, with its parameterization of the nonlinearity coefficient α and the dispersion coefficient β , was adopted from Martins (2023) and Martins et al. (2024). In the dataset generated for this study α and β are held equal, defining a one-parameter family of problems whose difficulty grows. Namely, for small $\alpha = \beta$ the dynamics are nearly linear and the solution is well-captured by low-frequency modes, while for large $\alpha = \beta$ the strong coupling between nonlinearity and dispersion produces sharper, more energetically distributed wave profiles that are fundamentally harder to learn. This controlled variation is precisely what makes the setting useful for comparing SciML architectures, with the goal of observing how each method learns the target solution’s frequency.

Classical numerical methods for the Boussinesq system rely on pseudospectral and finite-difference discretizations that exploit the periodic or nearly periodic spatial structure of the waves. Pseudospectral (**trefethen2000spectral**) schemes achieve spectral accuracy on the spatial derivatives and are therefore well suited to the dispersive character of the equations (ENG SIG-KARUP et al., 2012); this is the approach used to generate the reference dataset in this work. On the application side, operational coastal-engineering codes such as FUNWAVE (TODO, 2010) implement Boussinesq-type solvers for harbor and nearshore wave propagation, demonstrating the practical relevance of the model beyond the academic setting.

SciML methods have recently only been applied to the Boussinesq equation, not the specific Boussinesq System. Zheng et al. (2023) applied parameter-integrated PINNs with phase-domain decomposition to reconstruct two-dimensional Boussinesq dynamics including phase transitions and rogue-wave formation, identifying spectral bias as a key obstacle at high nonlinearity. Wang et al. (2024) trained a physics-informed network to infer velocity and pressure fields from surface-elevation data alone on a Boussinesq model, demonstrating the inverse-problem capabilities of the approach. Complementary studies on related nonlinear wave equations (ZHANG et al., 2024; KUMAR et al., 2025) have confirmed that purely data-driven and purely physics-driven networks each display characteristic failure modes at strong nonlinearity. Taken together, these results suggest that a systematic, parametric evaluation of FNO/PINO/PINN on the 1D Boussinesq system, which is the focus of this work, can be fruitful to understand better the interactions between the method’s ease of training and nonlinearities and dispersiveness.

1.3 Research Goals

This work implements and evaluates three SciML methods on the parametric 1D Boussinesq system: Physics-Informed Neural Networks (PINNs), Fourier Neural Operators (FNOs), and Physics-Informed Neural Operators (PINOs). All results are measured against a pseudospectral reference solver, which sets a high-accuracy baseline. The study is guided by the following questions.

- Accuracy relative to the pseudospectral baseline: how closely can each SciML method reproduce the reference solutions as nonlinearity and dispersion grow? Where does each method start to fail, and how does its error scale with the difficulty of the regime?
- Data-driven, physics-informed, and hybrid training: how does the presence or absence of supervised reference data affect the accuracy and training behavior of each method? Does adding a physics residual term improve generalization when data are scarce, or does it introduce new difficulties?
- Spectral bias across training regimes: how does spectral bias manifest in each architecture when the error spectrum is tracked band by band during training, and to what extent does the inclusion of data or physics constraints alter the frequency at which errors concentrate?
- The mechanism behind spectral bias: does Neural Tangent Kernel (NTK) theory genuinely predict the frequency imbalance seen on the Boussinesq system, or only describe it after the fact? We put the theory to two quantitative tests on the PINN residual, across a range of network widths. The first targets the frozen-kernel (lazy-training) assumption the analysis depends on: we measure how far the network parameters and the empirical kernel drift over training, and whether the kernel’s eigenvalue spectrum is essentially unchanged from initialization to convergence, as lazy training requires. The second targets the predicted driver of the bias: an eigenvalue spectrum that decays across many orders of magnitude, so that the modes carrying the high-frequency detail are learned far more slowly than the smooth, low-frequency ones. Confirming both would tie the frequency-band errors seen during training to a concrete mechanism rather than a loose analogy.

Together, these questions aim to provide a clear picture of the strengths and limitations of each SciML approach for dispersive, nonlinear wave problems, and to offer practical guidance for choosing among them.

1.4 Work Structure

This work is written with the goal of sharing SciML for senior undergraduates in mathematics, physics, or engineering. We recommend the reader to have some knowledge of scientific computing and linear algebra. The remaining chapters are organized as follows.

- Chapter 2 develops the mathematical background. It covers the Boussinesq system and its analytical properties, the pseudospectral solver, and the theoretical foundations of MLP, PINN, FNO, and PINO, including a discussion of spectral bias.
- Chapter 3 describes the experimental methodology. It details the dataset generation procedure, the model architectures, the training setup, and the evaluation metrics used throughout.
- Chapter 4 presents the numerical results and discusses the performance of each method across parameter regimes and spatial resolutions.
- Chapter 5 summarizes the main findings and outlines directions for future work.

2 MATHEMATICAL BACKGROUND

This chapter introduces the mathematical foundations of the methods studied in this work. Section 2.1 presents the Boussinesq system and the limiting wave-equation regime. Section 2.2 develops the pseudospectral numerical method used to generate reference solutions. Sections 2.3 and 2.4 introduce, respectively, the neural network and neural operator architectures studied in this work.

2.1 The Boussinesq System

The Boussinesq system is a model for long nonlinear dispersive waves in shallow water, governing the coupled evolution of two scalar fields: the free surface elevation $\eta(x, t)$ and the depth-averaged horizontal velocity $u(x, t)$, both depending on position x and time t . In the one-dimensional, spatially periodic setting studied in this work, the full system reads:

$$\begin{cases} \eta_t + u_x + \alpha (\eta u)_x = 0, \\ u_t - \frac{\beta}{3} u_{xxt} + \eta_x + \alpha u u_x = 0, \\ \eta(x, 0) = A \operatorname{sech}^2(x), \quad u(x, 0) = 0, \quad x \in [-L, L], \end{cases} \quad (2.1)$$

where $\alpha \geq 0$ is the nonlinearity parameter, $\beta > 0$ is the dispersion parameter, $A > 0$ is the initial amplitude, $L > 0$ is the domain half-length, and $\operatorname{sech}(z) = 2/(e^z + e^{-z})$ is the hyperbolic secant. The initial condition places a solitary-wave-like crest at the center $x = 0$ with zero initial velocity. This nondimensional form follows equation (3.1) of Martins (2023) (see also Martins et al. (2024)).

We note that $u u_x = \frac{1}{2}(u^2)_x$, so the momentum equation in (2.1) can equivalently be written with the conservative flux $\frac{\alpha}{2}(u^2)_x$ in place of $\alpha u u_x$. Both forms appear in the implementations: the PINN residual uses $u u_x$ directly via automatic differentiation, while the pseudospectral solver exploits $\frac{1}{2}(u^2)_x$ for the physical-space product, as detailed in Section 2.2.

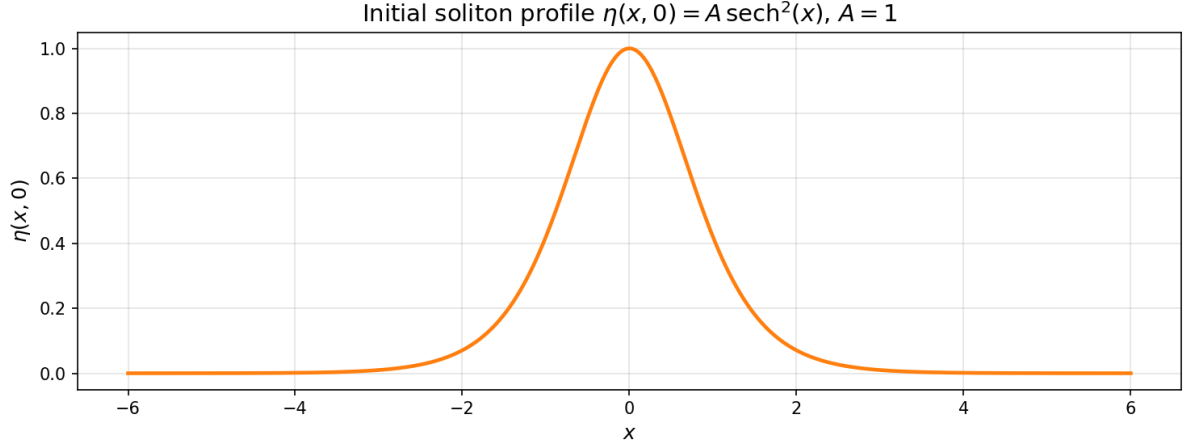


Figure 2.1 – Initial condition $\eta(x, 0) = A \operatorname{sech}^2(x)$ for $A = 1$.

Source: The author (2026).

2.1.1 Recovering the Wave Equation

In the doubly linear limit $\alpha = \beta = 0$, all nonlinear and dispersive terms vanish and the Boussinesq system reduces to the symmetric hyperbolic system:

$$\eta_t + u_x = 0, \quad (2.2)$$

$$u_t + \eta_x = 0. \quad (2.3)$$

To obtain a single scalar equation, differentiate (2.2) with respect to t :

$$\eta_{tt} + u_{xt} = 0. \quad (2.4)$$

Differentiate (2.3) with respect to x :

$$u_{tx} + \eta_{xx} = 0. \quad (2.5)$$

Since $u_{xt} = u_{tx}$, subtracting the second from the first yields the wave equation for η :

$$\eta_{tt} = \eta_{xx}, \quad (2.6)$$

and the reverse procedure gives the same equation for u .

D'Alembert's formula (DEBNATH, 2012) provides the general solution to (2.6). Given initial data

$$\eta(x, 0) = f(x), \quad \eta_t(x, 0) = g(x), \quad (2.7)$$

the solution is:

$$\eta(x, t) = \frac{f(x+t) + f(x-t)}{2} + \frac{1}{2} \int_{x-t}^{x+t} g(s) ds. \quad (2.8)$$

We now apply this to the initial conditions (2.1). From (2.2), $\eta_t(x, 0) = -u_x(x, 0) = 0$, so:

$$f(x) = A \operatorname{sech}^2(x), \quad g(x) = 0. \quad (2.9)$$

Since $g \equiv 0$, the integral in (2.8) vanishes and the surface elevation evolves as:

$$\eta(x, t) = \frac{A}{2} [\operatorname{sech}^2(x+t) + \operatorname{sech}^2(x-t)]. \quad (2.10)$$

For u , from (2.3) we have $u_t(x, 0) = -\eta_x(x, 0)$, giving:

$$f(x) = 0, \quad g(x) = -\eta_x(x, 0). \quad (2.11)$$

The $f = 0$ term vanishes in (2.8), leaving:

$$u(x, t) = \frac{1}{2} \int_{x-t}^{x+t} (-\eta_x(s, 0)) ds = -\frac{1}{2} [\eta(x+t, 0) - \eta(x-t, 0)]. \quad (2.12)$$

Substituting $\eta(x, 0) = A \operatorname{sech}^2(x)$:

$$u(x, t) = \frac{A}{2} [\operatorname{sech}^2(x-t) - \operatorname{sech}^2(x+t)]. \quad (2.13)$$

The initial bump of amplitude A splits into two counter-propagating pulses of amplitude $A/2$, traveling at speed ± 1 without changing shape. Any deviation from this wave-equation baseline in the full system (2.1) is attributable either to dispersion ($\beta > 0$) or to nonlinearity ($\alpha > 0$).

2.2 Pseudospectral Method

The pseudospectral method exploits two complementary representations of the solution: Fourier space, where derivatives become exact algebraic multiplications, and physical space, where pointwise nonlinear operations are inexpensive. By alternating between the two representations as needed, the method attains spectral accuracy in space while handling the nonlinear coupling efficiently.

2.2.1 Spatial Discretization and the Discrete Fourier Transform

The spatial domain $\Omega = [-L, L]$ of total length $2L$ is discretized into N_x equispaced points:

$$x_j = -L + j \Delta x, \quad \Delta x = \frac{2L}{N_x}, \quad j = 0, \dots, N_x - 1. \quad (2.14)$$

The Discrete Fourier Transform (DFT) of a function f sampled at these points is (TREFETHEN, 2000):

$$\hat{f}(k) = \sum_{j=0}^{N_x-1} f(x_j) e^{-2\pi i k j / N_x}, \quad k = -\frac{N_x}{2}, \dots, \frac{N_x}{2} - 1, \quad (2.15)$$

with its inverse recovering the physical values:

$$f(x_j) = \frac{1}{N_x} \sum_k \hat{f}(k) e^{2\pi i k j / N_x}. \quad (2.16)$$

The fundamental identity of the DFT with respect to differentiation is that, for a periodic function, the spatial derivative transforms into a pointwise multiplication (TREFETHEN, 2000; ENGSIG-KARUP et al., 2012):

$$\widehat{(f_x)}(k) = i \frac{\pi k}{L} \hat{f}(k), \quad \widehat{(f_{xx})}(k) = -\frac{\pi^2 k^2}{L^2} \hat{f}(k). \quad (2.17)$$

This is in contrast to finite-difference methods, where derivatives are approximated with truncation error; in the DFT, equation (2.17) is exact for periodic functions.

We define the wavenumber of mode k on $[-L, L]$ as:

$$\xi_k := \frac{\pi k}{L}. \quad (2.18)$$

With this definition, the derivative identities read:

$$\widehat{(f_x)}(k) = i \xi_k \hat{f}(k), \quad \widehat{(f_{xx})}(k) = -\xi_k^2 \hat{f}(k). \quad (2.19)$$

2.2.2 Spectral Formulation of the Boussinesq System

To bring the Boussinesq system (2.1) into the Fourier domain (MARTINS, 2023), we apply the DFT to each equation and use the derivative identities (2.19). The first equation, treating $\widehat{(\eta u)}(k, t)$ as a given nonlinear term, becomes:

$$\hat{\eta}_t(k, t) = -i \xi_k \hat{u}(k, t) - \alpha i \xi_k \widehat{(\eta u)}(k, t). \quad (2.20)$$

The second equation requires more care because of the u_{xxt} term. Since spatial derivatives commute with the DFT:

$$\widehat{(u_{xxt})}(k, t) = -\xi_k^2 \hat{u}_t(k, t). \quad (2.21)$$

Substituting into the DFT of the momentum equation in (2.1):

$$\hat{u}_t(k, t) + \frac{\beta}{3} \xi_k^2 \hat{u}_t(k, t) + i \xi_k \hat{\eta}(k, t) + \alpha i \xi_k \widehat{(\frac{1}{2}u^2)}(k, t) = 0. \quad (2.22)$$

Collecting the $\hat{u}_t(k, t)$ terms on the left and solving algebraically:

$$\hat{u}_t(k, t) = \frac{-i \xi_k \hat{\eta}(k, t) - \alpha i \xi_k \widehat{(\frac{1}{2}u^2)}(k, t)}{1 + \frac{\beta \xi_k^2}{3}}. \quad (2.23)$$

The operator $1 - \frac{\beta}{3} \frac{\partial^2}{\partial x^2}$, which requires a linear solve in physical space, becomes the diagonal scalar $1 + \frac{\beta \xi_k^2}{3}$ in Fourier space, reducing its inversion to a single division per mode.

Equations (2.20) and (2.23) can be written jointly as a first-order ODE system in time for each fixed wavenumber $k \in \mathbb{Z}$:

$$\begin{cases} \hat{\eta}_t(k, t) = -i \xi_k \hat{u}(k, t) - \alpha i \xi_k \widehat{(\eta u)}(k, t), \\ \hat{u}_t(k, t) = \frac{-i \xi_k \hat{\eta}(k, t) - \alpha i \xi_k \widehat{(\frac{1}{2}u^2)}(k, t)}{1 + \frac{\beta \xi_k^2}{3}}. \end{cases} \quad (2.24)$$

The PDE system (2.1), two equations in (x, t) , is thus equivalent to an infinite family of two-dimensional ODE systems (2.24), one for each $k \in \mathbb{Z}$, coupled through the nonlinear terms.

The nonlinear terms $\widehat{(\eta u)}(k, t)$ and $\widehat{(\frac{1}{2}u^2)}(k, t)$ cannot be computed directly in Fourier space without a convolution sum, which costs $O(N_x^2)$. The pseudospectral approach avoids this by computing products in physical space. For $\widehat{(\eta u)}(k, t)$, the procedure is:

1. recover $\eta(x_j)$ and $u(x_j)$ by applying the inverse DFT (2.16);
2. compute the product $(\eta u)_j = \eta(x_j) \cdot u(x_j)$ pointwise on the grid;
3. apply the DFT (2.15) to obtain $\widehat{(\eta u)}(k, t)$.

Similarly, $\widehat{(\frac{1}{2}u^2)}(k, t)$ is obtained by the same three-step procedure with $u^2(x_j) = u(x_j)^2$, where $u(x_j) = \text{IDFT}(\hat{u})(x_j)$; and $(\eta u)(x_j) = \text{IDFT}(\hat{\eta})(x_j) \cdot \text{IDFT}(\hat{u})(x_j)$. Since each DFT/IDFT costs $O(N_x \log N_x)$ via the Fast Fourier Transform (FFT), this pseudospectral strategy assembles each nonlinear term in $O(N_x \log N_x)$ rather than $O(N_x^2)$.

2.2.3 Time Evolution via Runge-Kutta 4

We define the pseudospectral right-hand side operator F as the map

$$F(\hat{\eta}, \hat{u})(k) = \left(\frac{-i \xi_k \hat{u}(k) - \alpha i \xi_k \widehat{(\eta u)}(k)}{1 + \frac{\beta \xi_k^2}{3}} \right), \quad (2.25)$$

so that $F(\hat{\eta}, \hat{u}) = (\hat{\eta}_t, \hat{u}_t)$ as given by equations (2.20)–(2.23), with nonlinear terms assembled by the pseudospectral procedure above. Time evolution uses the classical fourth-order Runge-Kutta (RK4) scheme (SÜLI et al., 2003; BURDEN et al., 2011). Given the spectral state $y_n = (\hat{\eta}^n, \hat{u}^n)$ at time t_n , the four stage evaluations $\kappa_1, \kappa_2, \kappa_3, \kappa_4$ and the update to $t_{n+1} = t_n + \Delta t$ are:

$$\begin{aligned} \kappa_1 &= F(y_n), \\ \kappa_2 &= F\left(y_n + \frac{\Delta t}{2} \kappa_1\right), \\ \kappa_3 &= F\left(y_n + \frac{\Delta t}{2} \kappa_2\right), \\ \kappa_4 &= F(y_n + \Delta t \kappa_3), \\ y_{n+1} &= y_n + \frac{\Delta t}{6} (\kappa_1 + 2 \kappa_2 + 2 \kappa_3 + \kappa_4). \end{aligned} \quad (2.26)$$

At each stage, the intermediate state is decoded to physical space to assemble the nonlinear products, then encoded back to Fourier space to evaluate the spectral derivatives. This split between physical-space products and Fourier-space derivatives is the defining operation of the pseudospectral method.

2.2.4 Training Dataset

The numerical experiments in this work vary only the PDE parameters (α, β) while keeping the initial condition fixed at (2.1). The models are therefore trained and evaluated on the dataset

$$\mathcal{S} = \left\{ ((\alpha_i, \beta_i), (\eta_i, u_i)) \right\}_{i=1}^M, \quad (2.27)$$

where (α_i, β_i) are the nonlinearity and dispersion parameters for the i -th sample, and (η_i, u_i) is the corresponding space-time solution obtained from the pseudospectral solver of Section 2.2 with those parameters and the shared initial condition (2.1).

2.3 Neural Networks

This section introduces the neural network architectures central to this work together with the theory that explains their training behavior. We begin with the Multilayer Perceptron as the building block, develop the Neural Tangent Kernel framework and the spectral bias it predicts for a network fitting data on a regular grid, and then turn to Physics-Informed Neural Networks, showing how the same kernel analysis extends to the physics-informed setting where spectral bias is a well-documented obstacle.

2.3.1 Multilayer Perceptrons

In this work, we focus on Multilayer Perceptrons (MLPs), popularized by Rumelhart et al. (1986). These networks are compositions of affine transformations parametrized by tunable weights and biases, separated by nonlinear activations, where an optimization process fits the network to a dataset.

Given a labeled dataset $\{(x_i, u_i)\}_{i=1}^N$, the objective is to fit the function $u_\theta(x)$ by minimizing a loss function, here the Mean Squared Error (MSE). With $\|\cdot\|$ denoting the Euclidean norm, the unconstrained optimization problem is:

$$\mathcal{L}_{\text{MLP}}(\theta) = \frac{1}{N} \sum_{i=1}^N \|u_\theta(x_i) - u_i\|^2. \quad (2.28)$$

The MLP $u_\theta : \mathbb{R}^{d_{\text{in}}} \rightarrow \mathbb{R}^{d_{\text{out}}}$ is defined by the composition:

$$h^{(0)} = x \in \mathbb{R}^{d_{\text{in}}}, \quad (2.29)$$

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)}) \in \mathbb{R}^{N_{\text{MLP}}}, \quad l = 1, \dots, L_{\text{MLP}} - 1, \quad (2.30)$$

$$u_\theta(x) = W^{(L_{\text{MLP}})}h^{(L_{\text{MLP}}-1)} + b^{(L_{\text{MLP}})} \in \mathbb{R}^{d_{\text{out}}}. \quad (2.31)$$

where:

- $L_{\text{MLP}} \in \mathbb{N}$ is the number of layers and $N_{\text{MLP}} \in \mathbb{N}$ the number of neurons per hidden layer;
- $W^{(l)} \in \mathbb{R}^{N_{\text{MLP}} \times N_{\text{MLP}}}$ and $b^{(l)} \in \mathbb{R}^{N_{\text{MLP}}}$ are the weight matrix and bias vector at layer l , with $W^{(1)} \in \mathbb{R}^{d_{\text{in}} \times N_{\text{MLP}}}$ and $W^{(L_{\text{MLP}})} \in \mathbb{R}^{N_{\text{MLP}} \times d_{\text{out}}}$ adapted at the boundaries;
- $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a sigmoidal activation function applied elementwise, satisfying $\sigma(t) \rightarrow 1$ as $t \rightarrow +\infty$ and $\sigma(t) \rightarrow 0$ as $t \rightarrow -\infty$;
- $\theta = \{W^{(l)}, b^{(l)}\}_{l=1}^{L_{\text{MLP}}}$ denotes all trainable parameters.

If the activation function is sigmoidal, MLPs are universal approximators: they can approximate any continuous function on a compact domain to arbitrary precision (CYBENKO, 1989).

In Figure 2.2, we show a common depiction of the neural network. Here the input vector is the pair $(x, t) \in \mathbb{R}^2$ and the output is $(\eta_\theta(x, t), u_\theta(x, t)) \in \mathbb{R}^2$.

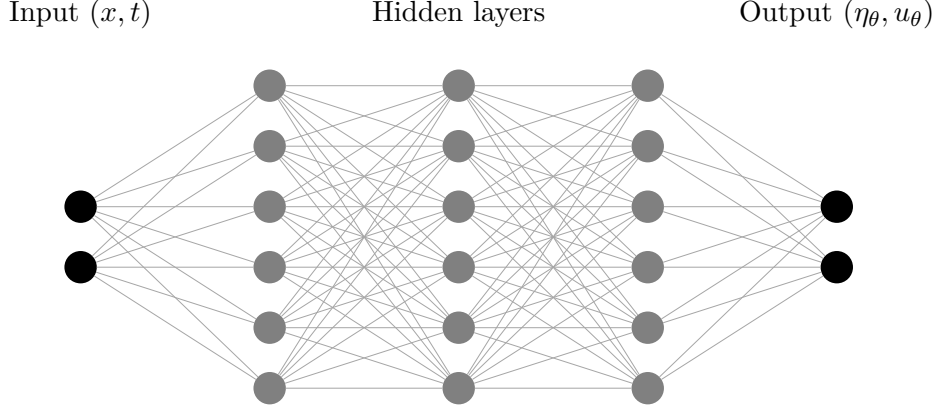


Figure 2.2 – Example of MLP architecture of input (x, t) , with $L_{\text{MLP}} = 3$ hidden layers and $N_{\text{MLP}} = 6$ neurons per layer.

Source: The author (2026).

2.3.2 Gradient Flow

To understand how an MLP minimizes its data-fitting loss (2.28), consider gradient descent with learning rate $\mu > 0$. Starting from an initialization $\theta^{(0)}$, each iteration updates the parameters by

$$\theta^{(k+1)} = \theta^{(k)} - \mu \nabla_{\theta} \mathcal{L}(\theta^{(k)}), \quad k = 0, 1, 2, \dots, \quad (2.32)$$

where $\theta^{(k)}$ denotes the parameter vector after k steps. Rearranging (2.32) and dividing by the step size μ isolates the per-step increment as a difference quotient:

$$\frac{\theta^{(k+1)} - \theta^{(k)}}{\mu} = -\nabla_{\theta} \mathcal{L}(\theta^{(k)}). \quad (2.33)$$

To pass from this discrete recursion to a continuous trajectory, assign to each iterate a training-time stamp

$$\tau^{(k)} = k\mu, \quad \text{so that} \quad \tau^{(k+1)} - \tau^{(k)} = \mu, \quad (2.34)$$

spacing successive iterates by μ along a time axis τ . We then regard the discrete sequence $\{\theta^{(k)}\}_{k=0}^{\infty}$ as samples of a smooth curve $\theta(\tau)$ evaluated at these instants:

$$\theta^{(k)} = \theta(\tau^{(k)}), \quad \theta^{(k+1)} = \theta(\tau^{(k+1)}) = \theta(\tau^{(k)} + \mu). \quad (2.35)$$

Substituting (2.35) into the left-hand side of (2.33) recasts it as a forward finite-difference quotient of θ in τ :

$$\frac{\theta(\tau^{(k)} + \mu) - \theta(\tau^{(k)})}{\mu} = -\nabla_{\theta} \mathcal{L}(\theta(\tau^{(k)})). \quad (2.36)$$

As the step size $\mu \rightarrow 0$, the spacing in (2.34) shrinks to zero, the discrete stamps $\tau^{(k)}$ fill out a continuous variable τ , and the left-hand side of (2.36) converges to the derivative $d\theta/d\tau$ by definition of the derivative. The recursion thus becomes the gradient flow ODE:

$$\frac{d\theta}{d\tau} = -\nabla_{\theta} \mathcal{L}(\theta), \quad \theta(0) = \theta^{(0)}. \quad (2.37)$$

The gradient flow (2.37) is a central object in optimization, and it is a key insight that many standard first-order methods can be recovered as different time-discretization schemes of this same continuous trajectory. The discrete update (2.32) is precisely the forward (explicit) Euler discretization of (2.37) with step size μ , so gradient descent is just one numerical integrator for the flow; applying instead the backward (implicit) Euler scheme recovers the proximal gradient method (PARIKH et al., 2013). Reading optimizers as discretizations of (2.37) thus organizes them into a single family, and it is the continuous flow that we analyze in what follows.

2.3.3 Neural Tangent Kernel

Consider the MLP fit to the dataset $\{(x_i, u_i)\}_{i=1}^N$ of Section 2.3.1. Group the pointwise fitting errors into a single error vector:

$$e(\theta) = \begin{bmatrix} u_{\theta}(x_1) - u_1 \\ \vdots \\ u_{\theta}(x_N) - u_N \end{bmatrix} \in \mathbb{R}^N. \quad (2.38)$$

Up to the constant factor $1/N$, the loss (2.28) is the squared norm of this vector, which we use in the form

$$\mathcal{L}(\theta) = \frac{1}{2} \|e(\theta)\|^2 = \frac{1}{2} \sum_{i=1}^N e_i(\theta)^2. \quad (2.39)$$

Define the Jacobian of the error vector with respect to the parameters:

$$J(\theta) = \frac{\partial e}{\partial \theta} \in \mathbb{R}^{N \times N_\theta}, \quad J_{ij} = \frac{\partial e_i}{\partial \theta_j}, \quad i = 1, \dots, N, \quad j = 1, \dots, N_\theta, \quad (2.40)$$

where N_θ is the total number of trainable parameters. Each row $J_i = (J_{i1}, \dots, J_{iN_\theta}) \in \mathbb{R}^{N_\theta}$ is the parameter-gradient of the i -th residual. Since $\mathcal{L} = \frac{1}{2} \sum_i e_i^2$, the j -th component of the loss gradient is:

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = \sum_{i=1}^N e_i(\theta) \frac{\partial e_i}{\partial \theta_j} = \sum_{i=1}^N J_{ij} e_i(\theta) = (J(\theta)^T e(\theta))_j. \quad (2.41)$$

Collecting all $j = 1, \dots, N_\theta$ into a column vector:

$$\nabla_\theta \mathcal{L} = J(\theta)^T e(\theta) \in \mathbb{R}^{N_\theta}. \quad (2.42)$$

Substituting (2.42) into the gradient flow (2.37):

$$\frac{d\theta}{d\tau} = -J(\theta)^T e(\tau). \quad (2.43)$$

To find how the error vector itself evolves, differentiate each component $e_i(\theta(\tau))$ with respect to τ by the chain rule:

$$\frac{de_i}{d\tau} = \sum_{j=1}^{N_\theta} \frac{\partial e_i}{\partial \theta_j} \frac{d\theta_j}{d\tau} = J_i(\theta) \cdot \frac{d\theta}{d\tau}. \quad (2.44)$$

Stacking this identity for all $i = 1, \dots, N$ into a matrix equation:

$$\frac{de}{d\tau} = J(\theta) \frac{d\theta}{d\tau}. \quad (2.45)$$

Substituting (2.43):

$$\frac{de}{d\tau} = J(\theta) (-J(\theta)^T e(\tau)) = - \underbrace{J(\theta) J(\theta)^T}_{K(\theta) \in \mathbb{R}^{N \times N}} e(\tau). \quad (2.46)$$

The matrix $K(\theta) = J(\theta)J(\theta)^T$ is the empirical Neural Tangent Kernel (NTK) (JACOT et al., 2018). It is symmetric positive semi-definite by construction: for any $v \in \mathbb{R}^N$,

$$v^T K(\theta) v = v^T J(\theta) J(\theta)^T v = \|J(\theta)^T v\|^2 \geq 0. \quad (2.47)$$

The (i, j) entry of $K(\theta)$ follows directly from expanding the matrix product $J(\theta)J(\theta)^T$:

$$K_{ij}(\theta) = \sum_{l=1}^{N_\theta} J_{il}(\theta) J_{jl}(\theta) = \sum_{l=1}^{N_\theta} \frac{\partial e_i}{\partial \theta_l}(\theta) \frac{\partial e_j}{\partial \theta_l}(\theta) = \nabla_{\theta} e_i(\theta)^T \nabla_{\theta} e_j(\theta). \quad (2.48)$$

This is the inner product in $\mathbb{R}^{N_\theta}$ of the parameter gradients of the network output at the i -th and j -th training points, where $e_i = u_\theta(x_i) - u_i$. Hence $K(\theta) \in \mathbb{R}^{N \times N}$ is the Gram matrix of the network's sensitivities across the whole dataset, and equation (2.46) shows that it alone governs how the fitting error evolves under gradient flow.

2.3.4 Spectral Bias

The spectral bias (or frequency principle) is the empirical observation that neural networks trained by gradient-based methods reduce the low-frequency components of the error faster than the high-frequency ones (RAHAMAN et al., 2019). This section derives that phenomenon directly from the NTK dynamics established above. When the training points lie on a regular grid and the target function is periodic, the error admits a discrete Fourier representation, and one can ask how its frequency content is approximated along the training iterations. We first establish the decay in the eigenbasis of the NTK and then, by transforming to Fourier space, show that it is precisely the low frequencies that are resolved first.

For a network with a sufficiently large number of parameters, the Jacobian $J(\theta)$ changes negligibly during training (JACOT et al., 2018; WANG; YU, et al., 2022). Writing $J_0 = J(\theta(0))$ and $K_0 = J_0 J_0^T$, the approximation $K(\theta(\tau)) \approx K_0$ turns equation (2.46) into a linear ODE with constant coefficients:

$$\frac{de}{d\tau} = -K_0 e(\tau), \quad e(0) = e_0. \quad (2.49)$$

Written component-wise, the i -th row of this system is

$$\frac{de_i}{d\tau} = - \sum_{j=1}^N K_0^{ij} e_j(\tau), \quad i = 1, \dots, N, \quad (2.50)$$

where $K_0^{ij} = \nabla_{\theta_0} e_i^T \nabla_{\theta_0} e_j$ is the inner product, at initialization, of the parameter gradients of the network output at training points x_i and x_j . The rate of change of the error at point i is therefore a weighted sum of all current errors, with weights given by these pairwise kernel values. This coupling is what makes the geometry of the sample grid and the architecture of the network jointly determine the convergence of every component.

Since $K_0 \in \mathbb{R}^{N \times N}$ is symmetric and positive semi-definite, the spectral theorem

guarantees a decomposition

$$K_0 = V \Lambda V^T, \quad (2.51)$$

where $V = [v_1 \mid \cdots \mid v_N]$ is an orthogonal matrix ($V^T V = I$) and

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N), \quad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0. \quad (2.52)$$

Introduce the change of coordinates

$$c(\tau) = V^T e(\tau), \quad (2.53)$$

so that, since V is orthogonal, $e(\tau) = V c(\tau)$. Differentiating (2.53) with respect to τ and substituting the ODE (2.49):

$$\frac{dc}{d\tau} = V^T \frac{de}{d\tau} = -V^T K_0 e(\tau). \quad (2.54)$$

Inserting $K_0 = V \Lambda V^T$ and $e(\tau) = V c(\tau)$:

$$\frac{dc}{d\tau} = -V^T (V \Lambda V^T) (V c(\tau)) = -(V^T V) \Lambda (V^T V) c(\tau) = -\Lambda c(\tau), \quad (2.55)$$

using $V^T V = I$ at each step. Since Λ is diagonal, this decouples into N independent scalar equations:

$$\frac{dc_k}{d\tau} = -\lambda_k c_k(\tau), \quad k = 1, \dots, N. \quad (2.56)$$

Each equation in (2.56) is a first-order linear ODE with constant coefficient (STROGATZ, 2018). Separating the variables and integrating both sides from the initial state at $\tau = 0$ up to time τ , with s and r as integration variables for the amplitude and the time,

$$\int_{c_k(0)}^{c_k(\tau)} \frac{ds}{s} = -\lambda_k \int_0^\tau dr \quad \implies \quad \ln \frac{|c_k(\tau)|}{|c_k(0)|} = -\lambda_k \tau. \quad (2.57)$$

Exponentiating both sides yields the solution

$$c_k(\tau) = c_k(0) e^{-\lambda_k \tau}, \quad (2.58)$$

where $c_k(0) = v_k^T e_0$ is the projection of the initial error e_0 onto the k -th eigenvector of K_0 . For $\lambda_k = 0$ the equation gives $c_k(\tau) = c_k(0)$ for all τ : that eigendirection never decays.

Returning to the original coordinates via $e(\tau) = V c(\tau)$:

$$e(\tau) = \sum_{k=1}^N (v_k^T e_0) e^{-\lambda_k \tau} v_k. \quad (2.59)$$

The error decomposes into N independent eigendirections. Each eigenvector v_k carries initial amplitude $v_k^T e_0$ and decays exponentially at rate λ_k : eigenvectors with large λ_k are suppressed quickly, while those with small λ_k persist throughout training. This per-eigendirection decay, each at the rate set by its eigenvalue, is the rigorous statement of spectral bias established by Cao et al. (2021); the same spectral ordering governs generalization as the training set grows (BORDELON et al., 2020).

Equation (2.59) is the continuous flow; gradient descent takes finite steps, and its error evolution inherits the same eigenstructure in discrete form. Linearizing the update (2.32) about the frozen Jacobian J_0 , a single step moves the parameters by $\theta^{(n+1)} - \theta^{(n)} = -\mu J_0^T e^{(n)}$, and to first order the error responds as $e^{(n+1)} - e^{(n)} = J_0 (\theta^{(n+1)} - \theta^{(n)})$. Combining the two,

$$e^{(n+1)} = e^{(n)} - \mu J_0 J_0^T e^{(n)} = (I - \mu K_0) e^{(n)}. \quad (2.60)$$

Iterating from $e^{(0)} = e_0$ and diagonalizing with $K_0 = V \Lambda V^T$,

$$e^{(n)} = (I - \mu K_0)^n e_0 = \sum_{k=1}^N (v_k^T e_0) (1 - \mu \lambda_k)^n v_k. \quad (2.61)$$

This is the forward-Euler counterpart of (2.59): each eigendirection is multiplied by $(1 - \mu \lambda_k)^n$ rather than $e^{-\lambda_k \tau}$, and the two coincide as $\mu \rightarrow 0$ with $\tau = n\mu$, since $(1 - \mu \lambda_k)^n \rightarrow e^{-\lambda_k \tau}$. The discrete mode contracts only when $|1 - \mu \lambda_k| < 1$, that is $\mu < 2/\lambda_k$, so the largest eigenvalue caps the stable learning rate at $\mu < 2/\lambda_{\max}$.

From (2.58), the amplitude of the k -th eigenvector at time τ satisfies $|c_k(\tau)| = |v_k^T e_0| e^{-\lambda_k \tau}$. To reduce this amplitude below a target precision $\varepsilon > 0$:

$$|v_k^T e_0| e^{-\lambda_k \tau} < \varepsilon \implies e^{-\lambda_k \tau} < \frac{\varepsilon}{|v_k^T e_0|} \implies -\lambda_k \tau < \ln \frac{\varepsilon}{|v_k^T e_0|} \implies \tau > \frac{1}{\lambda_k} \ln \frac{|v_k^T e_0|}{\varepsilon}. \quad (2.62)$$

The minimum training time to resolve the k -th eigenvector to precision ε is therefore

$$\tau_k = \frac{1}{\lambda_k} \ln \frac{|v_k^T e_0|}{\varepsilon}. \quad (2.63)$$

Since $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N \geq 0$, equation (2.63) gives $\tau_1 \leq \tau_2 \leq \dots \leq \tau_N$: higher-indexed eigenvectors require more training time. Arora et al. (2019) reached similar conclusions from a discrete-time analysis of gradient descent, showing that the convergence speed of each eigenvector is governed by its eigenvalue and by the projection of the target onto that eigenvector.

The decomposition (2.59) is expressed in the eigenbasis $\{v_k\}_{k=1}^N$ of the NTK, which

need not coincide with any physically meaningful basis. When the N training points lie on a regular grid and the target is periodic, however, the natural basis is the Fourier one, and recasting the dynamics there shows that the decay is genuinely organized by frequency. Let $F \in \mathbb{C}^{N \times N}$ be the Fourier matrix, that is, the discrete Fourier transform (DFT) matrix with entries

$$F_{jl} = \frac{1}{\sqrt{N}} \omega^{jl}, \quad \omega = e^{-2\pi i/N}, \quad j, l = 0, \dots, N-1. \quad (2.64)$$

With this normalization F is unitary, $F^*F = I$, so that $F^{-1} = F^*$, where F^* is the conjugate transpose (GOLUB et al., 2013; STRANG, 2006). Applying F to the error vector yields its spectrum, with inverse transform F^{-1} :

$$\hat{e}(\tau) = F e(\tau), \quad e(\tau) = F^{-1} \hat{e}(\tau) = F^* \hat{e}(\tau). \quad (2.65)$$

In particular $\hat{e}_0 = F e_0$ is the spectrum of the initial error, the difference between the spectra of the initial network output and the target. Since F is constant in τ , applying it to the linearized dynamics (2.49) carries the error evolution into the frequency domain:

$$\frac{d\hat{e}}{d\tau} = F \frac{de}{d\tau} = -F K_0 e(\tau) = -F K_0 F^{-1} \hat{e}(\tau). \quad (2.66)$$

Inserting the eigendecomposition $K_0 = V \Lambda V^T$ from (2.51),

$$F K_0 F^{-1} = F V \Lambda V^T F^{-1} = (FV) \Lambda (FV)^* = (FV) \Lambda (FV)^{-1}, \quad (2.67)$$

where $V^T F^{-1} = V^* F^* = (FV)^*$ uses that V is real orthogonal, and $(FV)^* = (FV)^{-1}$ uses that a product of unitary matrices is unitary. The columns of FV , namely the Fourier transforms $\{Fv_k\}_{k=1}^N$ of the NTK eigenvectors, therefore diagonalize the frequency-domain dynamics. Defining the modal amplitudes $\tilde{c}(\tau) = (FV)^{-1} \hat{e}(\tau)$ and repeating the steps (2.53)–(2.58) gives

$$\frac{d\tilde{c}_k}{d\tau} = -\lambda_k \tilde{c}_k(\tau) \quad \implies \quad \tilde{c}_k(\tau) = \tilde{c}_k(0) e^{-\lambda_k \tau}. \quad (2.68)$$

These are the same amplitudes as before: $\tilde{c} = (FV)^{-1} F e = V^{-1} e = V^T e = c$, the projection of the error onto the NTK eigenvectors, now read in Fourier space. Transforming back through (2.65) yields the evolution of the error spectrum,

$$\hat{e}(\tau) = \sum_{k=1}^N [(Fv_k)^* \hat{e}_0] e^{-\lambda_k \tau} (Fv_k). \quad (2.69)$$

Equation (2.69) is the spectral-bias statement in the literal sense of the frequency spectrum: the error spectrum is a superposition of the transformed eigenvectors Fv_k , each damped at its own rate λ_k . The frequencies resolved first are those carried by the eigenvectors with the largest eigenvalues; empirically these eigenvectors are smooth and concentrated in the low Fourier modes (RAHAMAN et al., 2019; XU et al., 2020), so the low-frequency content of the error decays first while the high-frequency content persists, exactly the behaviour observed in practice.

To close, note that when the network is initialized at small scale its output u_{θ_0} is negligible relative to the target, a common simplification in spectral-bias analyses. The initial error is then the target with opposite sign, and so is its spectrum:

$$\hat{e}_0 = F(u_{\theta_0} - u) = \hat{u}_{\theta_0} - \hat{u} \approx -\hat{u} = - \begin{bmatrix} \hat{u}(0) \\ \vdots \\ \hat{u}(N-1) \end{bmatrix}. \quad (2.70)$$

Substituting (2.70) into (2.69) casts the dynamics as a reconstruction of the target frequency by frequency: each Fourier mode of u is acquired at a rate set by the NTK eigenvalues, the smooth low-frequency content emerging first and the fine high-frequency detail last. This is exactly the regime anticipated at the start of the section — a periodic target on a regular grid — in which the quality of the approximation can be tracked mode by mode along the training iterations.

This last observation can be made quantitative. Taking the modulus of the modal amplitude (2.68) isolates the magnitude of the k -th component along training,

$$|\tilde{c}_k(\tau)| = |\tilde{c}_k(0)| e^{-\lambda_k \tau}, \quad |\tilde{c}_k(0)| = |(Fv_k)^* \hat{e}_0| \approx |(Fv_k)^* \hat{u}|, \quad (2.71)$$

where the last estimate uses the small-scale initialization (2.70), so that $|(Fv_k)^* \hat{u}|$ measures how strongly the target overlaps with the k -th eigen-mode. Training, however, advances in discrete steps: by the construction of the gradient flow (2.34), the continuous time and the iteration count n are related by $\tau = n\mu$, with μ the learning rate. Substituting $\tau = n\mu$ into (2.71) and requiring the amplitude to fall below a target precision $\varepsilon > 0$,

$$|(Fv_k)^* \hat{u}| e^{-\lambda_k n\mu} < \varepsilon \implies n > \frac{1}{\mu \lambda_k} \ln \frac{|(Fv_k)^* \hat{u}|}{\varepsilon}, \quad (2.72)$$

gives an estimate of the number of gradient-descent iterations needed to resolve the k -th frequency:

$$n_k = \left\lceil \frac{1}{\mu \lambda_k} \ln \frac{|(Fv_k)^* \hat{u}|}{\varepsilon} \right\rceil. \quad (2.73)$$

2.3.5 Physics-Informed Neural Networks

Introduced by Raissi et al. (2019), Physics-Informed Neural Networks (PINNs) insert the PDE directly into the optimization process, forcing the neural network to satisfy the model equations and approximating the PDE solution.

Consider the general PDE Initial Value Problem:

$$\begin{cases} \mathcal{D}[u](x, t) = 0, & \forall (x, t) \in \Omega \times (0, T], \\ u(x, 0) = u_0(x), & x \in \Omega, \end{cases} \quad (2.74)$$

where $\mathcal{D}$ is a differential operator on u , $\Omega \times (0, T]$ is the spatio-temporal domain, and u_0 is the initial condition.

The PDE residual loss penalizes violation of the governing equation at collocation points $\{(x_i, t_i)\}_{i=1}^{N_{\mathcal{D}}} \subset \Omega \times (0, T]$:

$$\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{N_{\mathcal{D}}} \sum_{i=1}^{N_{\mathcal{D}}} \|\mathcal{D}[u_{\theta}](x_i, t_i)\|^2. \quad (2.75)$$

The initial condition is enforced at collocation points $\{x_i\}_{i=1}^{N_0} \subset \Omega$:

$$\mathcal{L}_{\text{ic}}(\theta) = \frac{1}{N_0} \sum_{i=1}^{N_0} \|u_{\theta}(x_i, 0) - u_0(x_i)\|^2. \quad (2.76)$$

The combined PINN loss is:

$$\mathcal{L}_{\text{PINN}}(\theta) = \mathcal{L}_{\text{pde}}(\theta) + \mathcal{L}_{\text{ic}}(\theta). \quad (2.77)$$

Due to the compositional structure of MLPs, PINN implementations evaluate PDE residuals via Automatic Differentiation (AD) (LINNAINMAA, 1970; PASZKE et al., 2019), using ADAM (KINGMA et al., 2014) or L-BFGS (NOCEDAL, 1980) as the optimizer.

Application to the Boussinesq Dataset

For the Boussinesq system, a separate PINN is trained for each sample $(\alpha_i, \beta_i) \in \mathcal{S}$ (2.27). The differential operator $\mathcal{D}$ in the generic problem (2.74) is instantiated as the pair of Boussinesq residuals with fixed parameters (α_i, β_i) :

$$\mathcal{D}_1[u_{\theta}] = (\eta_{\theta})_t + (u_{\theta})_x + \alpha_i ((\eta_{\theta} u_{\theta})_x), \quad (2.78)$$

$$\mathcal{D}_2[u_{\theta}] = (u_{\theta})_t - \frac{\beta_i}{3} (u_{\theta})_{xxt} + (\eta_{\theta})_x + \alpha_i (u_{\theta} (u_{\theta})_x). \quad (2.79)$$

The combined loss (2.77) for sample i then reads:

$$\begin{aligned} \mathcal{L}_{\text{PINN}}^i(\theta) = & \frac{1}{N_{\mathcal{D}}} \sum_{l=1}^{N_{\mathcal{D}}} \left[\mathcal{D}_1[u_{\theta}](x_l, t_l)^2 + \mathcal{D}_2[u_{\theta}](x_l, t_l)^2 \right] \\ & + \frac{1}{N_0} \sum_{l=1}^{N_0} \left[(\eta_{\theta}(x_l, 0) - \eta_0(x_l))^2 + (u_{\theta}(x_l, 0) - u_0(x_l))^2 \right], \end{aligned} \quad (2.80)$$

where η_0 and u_0 are given by the shared initial condition (2.1). Because the parameters (α_i, β_i) enter the residuals explicitly, each PINN encodes knowledge of a single sample and must be retrained for every new parameter pair.

2.3.6 Spectral Bias in Physics-Informed Neural Networks

The Neural Tangent Kernel analysis of Sections 2.3.3–2.3.4 was carried out for an MLP fitting data on a uniform grid, the setting in which the kernel is approximately circulant and its eigenvectors are genuine Fourier modes. A PINN does not fit data directly: its loss (2.77) combines an initial-condition term with a PDE-residual term. The same machinery nevertheless applies once both constraints are stacked into a single error vector, now indexed over the N_r residual points in place of the N data points of the MLP,

$$e(\theta) = \begin{bmatrix} u_{\theta}(x_1, 0) - u_0(x_1) \\ \vdots \\ u_{\theta}(x_{N_0}, 0) - u_0(x_{N_0}) \\ \mathcal{D}[u_{\theta}](x_1, t_1) \\ \vdots \\ \mathcal{D}[u_{\theta}](x_{N_{\mathcal{D}}}, t_{N_{\mathcal{D}}}) \end{bmatrix} \in \mathbb{R}^{N_r}, \quad N_r = N_0 + N_{\mathcal{D}}, \quad (2.81)$$

so that, up to normalization, $\mathcal{L}_{\text{PINN}}$ reduces to the same quadratic form $\frac{1}{2}\|e\|^2$ analyzed for the MLP. With this definition the gradient-flow derivation leading to (2.46) carries over verbatim, with the Jacobian now also containing the parameter gradients of the PDE residuals, and the error still obeys the linear dynamics $\frac{de}{d\tau} = -K_0 e$, with convergence of each eigenvector governed by its eigenvalue exactly as in (2.63) (WANG; YU, et al., 2022).

This prediction is borne out empirically. Spectral bias is among the most consistently reported obstacles in PINN training: networks fit smooth, low-frequency solutions readily but stall on sharp gradients and high-frequency content (RAHAMAN et al., 2019; XU et al., 2020; WANG; TENG, et al., 2021), and the same pathology underlies several documented PINN failure modes (KRISHNAPRIYAN et al., 2021). Wang, Yu, et al. (2022) traced the effect directly to the NTK, showing that its spectrum is concentrated at

low eigenvalues, so that high-frequency PDE residuals are the last to be minimized; for the Boussinesq system specifically, Wang et al. (2024) report the same difficulty in resolving the finer wave structure. A range of remedies target precisely this imbalance, including adaptive loss weighting (WANG; TENG, et al., 2021), Fourier feature embeddings, and domain decomposition. Understanding the spectral bias is therefore essential background for the neural-operator methods studied in the remainder of this work, whose ability to capture the high-frequency content of the Boussinesq solutions is examined directly in the numerical results of Chapter 4.

2.4 Neural Operators

Neural operators generalize neural networks from learning finite-dimensional mappings to learning operators between infinite-dimensional function spaces. This section introduces the general framework, then the Fourier Neural Operator (FNO) as the principal architecture, and finally the Physics-Informed Neural Operator (PINO) as the physics-constrained extension.

2.4.1 General Definition

Let $D \subset \mathbb{R}^d$ be a bounded open domain and let $\mathcal{A}(D; \mathbb{R}^{d_a})$ and $\mathcal{U}(D; \mathbb{R}^{d_u})$ be Banach spaces of functions (KREYSZIG, 1978). The target operator is a (potentially nonlinear) map

$$G^\dagger : \mathcal{A} \rightarrow \mathcal{U}, \quad a \mapsto u = G^\dagger(a), \quad (2.82)$$

where both the input a and the output u are functions defined on D .

In this work, $G^\dagger$ is the solution operator of the Boussinesq system: given a parameter encoding (c, u_0) containing the PDE coefficients (α, β) and the initial condition u_0 , the operator returns the full space-time solution (η, u) .

A neural operator G_θ approximates $G^\dagger$ by composing linear integral operators with pointwise nonlinearities. Each linear layer applies an operator of the form:

$$(\mathcal{K}v)(x) = \int_D \kappa(x, y) v(y) dy, \quad (2.83)$$

where $\kappa : D \times D \rightarrow \mathbb{R}^{d_u \times d_u}$ is a learned kernel. The full neural operator architecture stacks these integral layers into the composition:

$$G_\theta = Q \circ \sigma(W_L + \mathcal{K}_L) \circ \cdots \circ \sigma(W_1 + \mathcal{K}_1) \circ P, \quad (2.84)$$

where:

- $P : \mathbb{R}^{d_a} \rightarrow \mathbb{R}^{d_c}$ is the lifting operator, mapping the input $a(x)$ pointwise to a higher-dimensional channel representation with $d_c \gg d_a$;
- $\sigma(W_\ell + \mathcal{K}_\ell)$, for $\ell = 1, \dots, L$, are the operator layers, each combining a local linear map W_ℓ with a nonlocal integral operator $\mathcal{K}_\ell$, followed by a pointwise activation σ ;
- $Q : \mathbb{R}^{d_c} \rightarrow \mathbb{R}^{d_u}$ is the projection operator mapping the final channel representation back to the output dimension.

Different choices of kernel parametrization for each $\mathcal{K}_\ell$ give rise to different neural operator architectures.

2.4.2 Fourier Neural Operators

Presented by Li et al. (2020), the Fourier Neural Operator (FNO) is the neural operator (2.84) in which each $\mathcal{K}_\ell$ uses a shift-invariant kernel: $\kappa_\ell(x, y) = \kappa_\ell(x - y)$. This assumption turns each integral operator into a convolution, which by the convolution theorem can be evaluated efficiently in the frequency domain.

Each spectral layer updates the channel representation via:

$$v_{\ell+1}(x) = \sigma\left((W_\ell v_\ell)(x) + (\mathcal{K}_\ell v_\ell)(x)\right). \quad (2.85)$$

The operator $\mathcal{K}_\ell$ acts via convolution with the shift-invariant kernel κ_ℓ :

$$(\mathcal{K}_\ell v_\ell)(x) = \int_D \kappa_\ell(x - y) v_\ell(y) dy. \quad (2.86)$$

By the convolution theorem, this equals:

$$(\mathcal{K}_\ell v_\ell)(x) = \mathcal{F}^{-1}\left(\mathcal{F}(\kappa_\ell) \cdot \mathcal{F}(v_\ell)\right)(x). \quad (2.87)$$

To see how this is parametrized, recall that for a periodic function f on $[0, 2\pi]$:

$$f(x) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{ikx}, \quad \hat{f}(k) = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-ikx} dx. \quad (2.88)$$

The convolution $(\kappa * v)$ simplifies to:

$$(\kappa * v)(x) = \int_0^{2\pi} \kappa(x - y) v(y) dy = \sum_{k \in \mathbb{Z}} (2\pi \hat{\kappa}(k) \hat{v}(k)) e^{ikx}. \quad (2.89)$$

Instead of learning κ_ℓ as a function in physical space, the FNO directly parametrizes its Fourier coefficients as learnable complex-valued matrices $R_{\ell,k} \approx 2\pi \hat{\kappa}_\ell(k)$. Only the

$|k| \leq k_{\max}$ low-frequency modes are retained, with all others set to zero; $k_{\max}$ is a hyperparameter.

Training Objective

The FNO is trained as a pure supervised learning problem. Given the dataset $\mathcal{D}$ (2.27) of parameter-solution pairs, the training objective minimizes the mean squared error between the predicted and reference space-time fields:

$$\mathcal{L}_{\text{FNO}}(\theta) = \frac{1}{M} \sum_{i=1}^M \frac{1}{N_x N_t} \sum_{j=0}^{N_x-1} \sum_{n=0}^{N_t} \left[(\eta_{\theta}^i(x_j, t_n) - \eta_i(x_j, t_n))^2 + (u_{\theta}^i(x_j, t_n) - u_i(x_j, t_n))^2 \right], \quad (2.90)$$

where $(\eta_{\theta}^i, u_{\theta}^i) := G_{\theta}(\alpha_i, \beta_i)$ is the network prediction for sample i .

Application to Non-Time-Periodic Problems

When the solution is also assumed to be periodic in time with period T (so that $D = [0, 2\pi] \times [0, T]$), shift-invariance extends to the time direction as well: $\kappa_{\ell}(x, y, t, \tau) = \kappa_{\ell}(x - y, t - \tau)$. The convolution then generalizes to:

$$(\mathcal{K}_{\ell} v_{\ell})(x, t) = \int_D \kappa_{\ell}(x - y, t - \tau) v_{\ell}(y, \tau) dy d\tau = \mathcal{F}^{-1} \left(\mathcal{F}(\kappa_{\ell}) \cdot \mathcal{F}(v_{\ell}) \right)(x, t), \quad (2.91)$$

and the FNO parametrizes the two-dimensional Fourier modes (k_x, k_t) with $|k_x| \leq k_{x,\max}$ and $|k_t| \leq k_{t,\max}$, both hyperparameters.

For problems that are not time-periodic, however, applying the Fourier transform along the temporal axis implicitly extends the signal periodically, potentially introducing a jump discontinuity at the temporal boundary. This can trigger the Gibbs phenomenon (GIBBS, 1899), producing oscillatory artifacts near the boundary that persist regardless of how many temporal modes are retained. Standard signal-processing techniques such as zero-padding and window functions (e.g. Hann, Hamming windows) can partially mitigate these effects, at the cost of altering the temporal structure of the signal. Li et al. (2020) acknowledge this limitation and provide an alternative formulation in which the Fourier layers operate only along the spatial axis while the model advances forward in time, avoiding the periodicity assumption in the temporal direction entirely. In this work we deliberately retain the two-dimensional space-time formulation, in which the Fourier layers act on both the spatial and the temporal axes, so that a single operator emits the entire trajectory in one forward pass. This keeps the architecture simple and uniform across space and time, but, as just noted, it reintroduces the temporal-periodicity assumption for a Boussinesq solution that is not time-periodic, and therefore admits a Gibbs artifact

at the temporal boundary $t = T$. Rather than suppress this artifact with windowing, we retain the plain formulation and examine the boundary behaviour directly in Chapter 4, where it turns out to distinguish the purely data-driven operator from its physics-informed variant.

2.4.3 Physics-Informed Neural Operators

Analogously to how PINNs add a physics-informed residual loss to the MLP training objective, the Physics-Informed Neural Operator (PINO) (LI et al., 2023a) incorporates PDE constraints into the FNO training. The FNO backbone remains unchanged; what differs is the loss function, which now combines a data term with residuals evaluated on the predicted space-time fields.

Unlike a PINN, which represents the solution as a single continuous function and differentiates it via automatic differentiation, the FNO outputs (η_θ, u_θ) as a discrete $N_x \times N_t$ array in a single forward pass over the full space-time domain. Evaluating the PDE residuals therefore reduces to numerically differentiating this output array.

Spatial derivatives are computed pseudospectrally using ξ_k (2.18): the DFT of η_θ along the spatial axis at each time t_n yields $\hat{\eta}_\theta(k, t_n)$, from which

$$(\eta_x)_j = \mathcal{F}^{-1}(i \xi_k \hat{\eta}_\theta(k, t_n))_j, \quad (\eta_{xx})_j = \mathcal{F}^{-1}(-\xi_k^2 \hat{\eta}_\theta(k, t_n))_j, \quad (2.92)$$

and identically for u_θ . Time derivatives are computed by finite differences: second-order central in the interior and first-order one-sided at the boundaries,

$$(\eta_\theta)_t^n = \begin{cases} \frac{\eta_\theta^{n+1} - \eta_\theta^n}{\Delta t}, & n = 0, \\ \frac{\eta_\theta^{n+1} - \eta_\theta^{n-1}}{2 \Delta t}, & 1 \leq n \leq N_t - 1, \\ \frac{\eta_\theta^n - \eta_\theta^{n-1}}{\Delta t}, & n = N_t, \end{cases} \quad (2.93)$$

and identically for $(u_\theta)_t^n$. The mixed term $(u_\theta)_{xxt}$ is obtained by first computing $(u_\theta)_{xx}$ spectrally and then applying the time-difference stencil to the result. This combination of spectral spatial derivatives and finite-difference temporal derivatives follows the reference implementation of Li et al. (2023a) (LI et al., 2023b).

With all derivatives available, the Boussinesq equations (2.1) define two differential operators acting on the predicted fields:

$$\mathcal{D}_1(\eta_\theta, u_\theta) := (\eta_\theta)_t + (u_\theta)_x + \alpha (\eta_\theta u_\theta)_x, \quad (2.94)$$

$$\mathcal{D}_2(\eta_\theta, u_\theta) := (u_\theta)_t - \frac{\beta}{3}(u_\theta)_{xxt} + (\eta_\theta)_x + \alpha (u_\theta(u_\theta)_x). \quad (2.95)$$

Here $(\eta_\theta^i, u_\theta^i) := G_\theta(\alpha_i, \beta_i)$ denotes the network output for sample i , and the subscript j,n indicates evaluation at the grid point (x_j, t_n) . The PINO loss is a weighted sum of three terms:

$$\mathcal{L}_{\text{PINO}}(\theta) = \lambda_{\text{pde}} \mathcal{L}_{\text{pde}}(\theta) + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}}(\theta) + \lambda_{\text{data}} \mathcal{L}_{\text{data}}(\theta), \quad (2.96)$$

where:

- $\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{M} \sum_{i=1}^M \frac{1}{N_x N_t} \sum_{j=0}^{N_x-1} \sum_{n=0}^{N_t} \left[\mathcal{D}_1(\eta_\theta^i, u_\theta^i)_{j,n}^2 + \mathcal{D}_2(\eta_\theta^i, u_\theta^i)_{j,n}^2 \right]$ penalizes violation of the Boussinesq equations at each grid point, with residuals evaluated using the parameters (α_i, β_i) of each sample;
- $\mathcal{L}_{\text{ic}}(\theta) = \frac{1}{M} \sum_{i=1}^M \frac{1}{N_x} \sum_{j=0}^{N_x-1} \left[(\eta_\theta(x_j, 0) - \eta_0(x_j))^2 + (u_\theta(x_j, 0) - u_0(x_j))^2 \right]$ enforces the shared initial condition (2.1);
- $\mathcal{L}_{\text{data}}(\theta) = \frac{1}{M} \sum_{i=1}^M \frac{1}{N_x N_t} \sum_{j=0}^{N_x-1} \sum_{n=0}^{N_t} \left[(\eta_\theta^i(x_j, t_n) - \eta_i(x_j, t_n))^2 + (u_\theta^i(x_j, t_n) - u_i(x_j, t_n))^2 \right]$ measures deviation from the reference solutions $(\eta_i, u_i) \in \mathcal{D}$ (2.27);
- $\lambda_{\text{pde}}, \lambda_{\text{ic}}, \lambda_{\text{data}} > 0$ are weighting hyperparameters.

The PINO therefore generalizes the FNO by adding physics supervision: it can be trained with fewer labeled samples by relying on the PDE residual for additional guidance.

3 METHODOLOGY

The theory of Chapter 2 becomes a set of runnable experiments here, and this chapter describes how they were built and read. Its layout follows the life of a single experiment. Everything downstream rests on three shared foundations, so we lay those first: the computing environment and the reproducibility rules that let any result be regenerated (Section 3.1), the pseudospectral data that serve at once as training targets and as ground truth (Section 3.2), and the error metrics that score every model on the same scale (Section 3.3). With the foundations in place, Section 3.4 presents the six models as a deliberate progression, each rung adding one ingredient to the one below it, from a plain data regressor up to a physics-constrained operator over the whole family. Section 3.5 then turns to the one behaviour every rung shares, spectral bias, and sets up two probes of its mechanism. Section 3.6 collects every hyperparameter into a single table.

3.1 Setup, Solver, and Reproducibility

Every piece of software behind these experiments was written for this work rather than taken from an existing package. The code runs on Python 3.11 with PyTorch 2.12 and CUDA 13.0 (PASZKE et al., 2019), and all timings come from one NVIDIA GeForce GTX 1660 SUPER card with 6 GB of memory. NumPy handles array bookkeeping outside the training loops, and Matplotlib and Plotly draw the figures. We used no pre-trained weights, no third-party operator-learning library and no external dataset. The pseudospectral solver, the multilayer perceptron, the Fourier layers and each optimization loop are our own, which is what let us instrument them for the spectral diagnostics that follow in Chapter 4. The complete implementation is available at <https://github.com/samuelkutz/TCC>.

Each learned architecture tracks its original reference as closely as we could manage. The Fourier Neural Operator follows Li et al. (2020, 2021), the Physics-Informed Neural Operator follows Li et al. (2023a) and its released code (LI et al., 2023b), and the Physics-Informed Neural Network follows Raissi et al. (2019). Every departure from a reference is flagged at the point where it is made.

Two conventions make the results reproducible. A single global seed of 37 initializes the Python, NumPy and PyTorch generators at the start of each run, and the cuDNN backend runs in deterministic mode with autotuning disabled. The spatial and temporal domains stay fixed throughout at

$$x \in [-L, L], \quad L = 60, \quad t \in [0, T], \quad T = 30, \quad (3.1)$$

and every experiment shares the initial condition of Equation (2.1) at unit amplitude $A = 1$, so that $\eta(x, 0) = \text{sech}^2(x)$ and $u(x, 0) = 0$. A unit-height, unit-width crest sits at the centre of a domain 120 units across, wide enough that the two counter-propagating pulses of the wave-equation limit (2.10)–(2.13) never reach the periodic boundary within the simulated horizon. Table 3.1 collects the simulation, dataset and evaluation constants that hold everywhere; each is restated below at the point where it first matters.

Table 3.1 – Fixed simulation, dataset and evaluation parameters shared by every experiment. Unless a section states otherwise, these values hold throughout.

Parameter	Value
<i>Domain and initial condition</i>	
Spatial half-length L ($\Omega = [-L, L]$)	60
Final time T ($t \in [0, T]$)	30
Initial wave amplitude A	1
Initial condition	$\eta(x, 0) = \text{sech}^2(x), u(x, 0) = 0$
<i>Discretization (per solve)</i>	
Spatial grid points N_x	128
Time levels N_t (127 RK4 steps)	128
Spatial spacing $\Delta x = 2L/N_x$	0.9375
Temporal spacing $\Delta t = T/(N_t - 1)$	≈ 0.2362
<i>Parameter sampling</i>	
Coupled nonlinearity and dispersion	$\alpha = \beta$
Number of training samples M	20
Sampling grid	<code>linspace(0.1, 4.0, 20)</code>
<i>Evaluation and reproducibility</i>	
Evaluation parameters $\alpha = \beta$	{0.1, 3.21, 4.2}
Evaluation resolutions N	{128, 256, 512}
Spectral evaluation resolution	256
Global random seed	37

Source: The author (2026).

3.2 Building the Parametric Dataset

The reference solutions carry two jobs at once. They are the supervised targets the data-driven regimes learn from, and they are the ground truth every model is later

scored against. The pseudospectral Runge–Kutta solver of Section 2.2 produces both, and its spectral spatial accuracy is what a dispersive wave problem needs from a high-fidelity baseline (ENGSIK-KARUP et al., 2012; TREFETHEN, 2000).

The parameter family is one-dimensional by design. Following Section 2.2.4, we hold the initial condition fixed and vary only the PDE coefficients, and we tie them together as $\alpha = \beta$ so that the two-parameter Boussinesq family collapses to a single axis of difficulty. Near the small end of that axis the dynamics stay close to linear and the solution energy occupies a handful of low modes; near the large end, steepening and dispersion combine to sharpen the crests and to spread their energy across a wider band of wavenumbers. We sample the axis uniformly at $M = 20$ points,

$$\alpha_i = \beta_i \in \text{linspace}(0.1, 4.0, 20), \quad i = 1, \dots, 20, \quad (3.2)$$

running from the near-linear value 0.1 up to the strongly nonlinear value 4.0.

Each parameter in (3.2) is fed to the solver on $N_x = 128$ spatial points and marched through $N_t = 128$ time levels, that is 127 Runge–Kutta steps over $[0, T]$, with spacings

$$\Delta x = \frac{2L}{N_x} = \frac{120}{128} = 0.9375, \quad \Delta t = \frac{T}{N_t - 1} = \frac{30}{127} \approx 0.2362. \quad (3.3)$$

A solve returns the two space-time fields $\eta_i(x_j, t_n)$ and $u_i(x_j, t_n)$ on the 128×128 grid. We store them in the channel-first tensors the two-dimensional operators expect, four channels in and two out:

$$X \in \mathbb{R}^{M \times 4 \times N_x \times N_t}, \quad Y \in \mathbb{R}^{M \times 2 \times N_x \times N_t}. \quad (3.4)$$

The four input channels hold, at every grid point, the initial elevation $\eta_i(x_j, 0)$, the initial velocity $u_i(x_j, 0)$ and the two constant coefficient fields α_i and β_i , with the initial-condition channels broadcast along the time axis so that the operator receives a full space-time array. The two output channels hold the reference fields η_i and u_i . From here on every panel reports the surface elevation η alone, since it carries the wave structure of interest and the velocity field behaves the same way.

These channels mix quantities of unlike magnitude: an elevation of order one, a velocity field, and coefficients that span more than an order of magnitude. To place them on a common footing we map every tensor into the unit interval with one affine, channel-wise min–max transform computed once over the whole dataset,

$$\tilde{v} = \frac{v - v_{\min}}{v_{\max} - v_{\min} + \varepsilon}, \quad \varepsilon = 10^{-12}, \quad (3.5)$$

where $v_{\min}$ and $v_{\max}$ are taken per channel across the sample, spatial and temporal axes. Operators train and predict in this normalized space, and their outputs return to physical units by inverting (3.5) before any metric or residual is formed. One consequence shapes how the physics-only regimes of Section 3.4.4 must be read: because $v_{\min}$ and $v_{\max}$ come from the reference solutions, even a model trained without a data-fitting term still inherits the output scale of the dataset through (3.5). “No data” therefore means no supervised residual on the interior fields, and it stops short of full independence from the reference.

3.3 A Common Yardstick

A fair comparison needs one ruler that every model is held against, so that a measured difference reflects the architecture and not the measurement. Every method is scored against a freshly recomputed pseudospectral reference with the shared metrics defined below. Three parameter values sample the difficulty axis,

$$\alpha = \beta \in \{0.1, 3.21, 4.2\}, \quad (3.6)$$

one near-linear, one intermediate and one strongly nonlinear. The middle value 3.21 is our default whenever a single representative parameter is called for, and it is where the single-parameter PINNs of Section 3.4.2 are trained. To probe the discretization-invariance operators are meant to enjoy, each trained operator is also queried, with no retraining, on grids of resolution

$$N \in \{128, 256, 512\}, \quad (3.7)$$

the training resolution together with two and four times finer. For each such query we resample the input channels of Section 3.2 onto the finer grid and recompute the reference at the matching resolution.

Let η_{true} and η_{pred} be the reference and predicted elevation on a common space-time grid. We report three complementary quantities.

Time-resolved relative error.

At each time level t_n we form the spatial L^2 discrepancy and divide it by the norm of the reference at that instant,

$$\mathcal{E}(t_n) = \frac{\|\eta_{\text{pred}}(\cdot, t_n) - \eta_{\text{true}}(\cdot, t_n)\|_2}{\|\eta_{\text{true}}(\cdot, t_n)\|_2}. \quad (3.8)$$

Read against time, $\mathcal{E}(t_n)$ locates where along the trajectory the error builds up, and its spread over all time levels is summarized by a box plot. This is the quantity behind the parameter and resolution panels of Chapter 4.

Spectral relative error.

To see *which* spatial frequencies are wrong, we take the spatial real DFT at a fixed time and compare amplitudes mode by mode. Writing $\hat{\eta}(k, t_n)$ for the amplitude at wavenumber index k , the per-mode relative error is

$$\mathcal{E}_{\text{spec}}(k, t_n) = \frac{||\hat{\eta}_{\text{pred}}(k, t_n)| - |\hat{\eta}_{\text{true}}(k, t_n)||}{\max(|\hat{\eta}_{\text{true}}(k, t_n)|, \delta)}, \quad (3.9)$$

where the floor δ , set proportional to the peak amplitude, keeps the ratio finite at modes whose reference amplitude is near zero. Averaging over the retained modes yields one spectral number per time level, and the spectral panels display both the mode-resolved curves at $t = 0$, $t = T/2$ and $t = T$ and the time evolution of the average.

Frequency-band error during training.

To watch spectral bias form as the optimizer runs, we split the spatial spectrum of the error into three contiguous wavenumber bands and log each band error every 500 iterations. The tracked ratio differs slightly from (3.9): here we apply the spatial real DFT to the *error field* $\eta_{\text{pred}} - \eta_{\text{true}}$ and to the reference, average each amplitude over the time levels, and take the quotient,

$$\mathcal{E}_{\text{band}}(k) = \frac{\langle |\widehat{(\eta_{\text{pred}} - \eta_{\text{true}})}(k, t_n)| \rangle_{t_n}}{\langle |\hat{\eta}_{\text{true}}(k, t_n)| \rangle_{t_n} + \varepsilon}, \quad \varepsilon = 10^{-12}. \quad (3.10)$$

Transforming the error before taking magnitudes makes this metric sensitive to the phase of the error, rather than to a mode-by-mode magnitude comparison. With $N_k = \lfloor N_x/2 \rfloor + 1$ the number of retained real-DFT modes, the low, mid and high bands cut the index range into quarters,

$$k < \frac{N_k}{4}, \quad \frac{N_k}{4} \leq k < \frac{N_k}{2}, \quad k \geq \frac{N_k}{2}, \quad (3.11)$$

which fall near $N_x/8$ and $N_x/4$ on the full grid. Each band error is the average of (3.10) over the modes in that band. These curves feed the band-tracking comparison of Chapter 4. One asymmetry in the logging matters for how the curves are read: each operator is tracked on the first dataset sample, $\alpha = \beta = 0.1$, while the single-parameter MLP and PINN are tracked on their own training solution at $\alpha = \beta = 3.21$. We keep this mismatch

in reference sample in mind when reading the results.

3.4 A Ladder of Approximators

The six models form a ladder rather than a list. Each rung keeps the machinery of the rung below and adds exactly one ingredient, so that any change in behaviour can be traced to that single addition. The first rung is a plain regressor that knows only the data. The second wraps the same network in the governing equation. The third lifts the problem from one solution to the whole family. The fourth adds the equation to that family-wide operator. Climbing the ladder in this order isolates, one step at a time, what physics contributes, what operator learning contributes, and what the two contribute together.

3.4.1 The Plain Regressor: MLP

The bottom rung is a Multilayer Perceptron that fits one solution from data alone. It is the fully connected network of Section 2.3.1, sending a space-time point to an elevation, $(x, t) \mapsto \eta_\theta(x, t)$, with two input units, one output unit, four hidden layers of 256 neurons and hyperbolic-tangent activations, for 198,401 trainable parameters. This same backbone is what the PINN of the next rung reuses, so any difference between them will come from the training signal and not the architecture.

Training is pure supervised regression. The objective is the mean-squared error (2.28) between the network output and the pseudospectral reference at the fixed parameter $\alpha = \beta = 3.21$, on the training grid of resolution 128. We optimize with Adam (KINGMA et al., 2014) at learning rate 10^{-3} for 15,000 iterations, taking one full-batch gradient step over the entire reference field per iteration. This matches the PINN budget exactly.

The value of this rung lies in what its objective omits. It carries no PDE-residual term and no initial-condition term, so the equation (2.1) never enters and the network sees only data. That makes it the cleanest testbed for spectral bias in this work: with nothing competing against the data fit, the order in which the network resolves the spatial frequencies of the target is set by the fitting dynamics alone. Any low-first, high-last ordering it shows is then spectral bias free of confounders. We measure that ordering with the frequency-band error (3.10)–(3.11), tracked on the network’s own training solution at $\alpha = \beta = 3.21$, and we tie it to the eigenvalue mechanism through the diagnostics of Section 3.5.

3.4.2 Adding the Equation: PINN

The second rung keeps the feed-forward backbone and adds the Boussinesq residual to its loss. The PINN of Section 2.3.5 carries the solution as one continuous map $(x, t) \mapsto (\eta_\theta, u_\theta)$ and is therefore tied to a single parameter pair. Each network is fully connected with two input units, two output units, four hidden layers of 256 neurons and hyperbolic-tangent activations, for 198,658 trainable parameters, and it is optimized with Adam (KINGMA et al., 2014) at learning rate 10^{-3} for 15,000 iterations.

The composite objective (2.80) has three parts. The residual term is evaluated at 5,000 interior collocation points resampled uniformly at random over $[-L, L] \times [0, T]$ at every iteration, and the derivatives entering the Boussinesq operators $\mathcal{D}_1, \mathcal{D}_2$ come from automatic differentiation, so they carry no discretization error. The initial-condition term is enforced at 500 equispaced points at $t = 0$. The data term, when present, is a supervised mean-squared error against a subsample of the pseudospectral reference. That last term is the one ingredient we toggle, giving two regimes:

- **PINN (no data)**: purely physics-informed, data weight 0; the network sees only the equation and the initial condition.
- **PINN (with data)**: hybrid, data weight 1; the network additionally fits the reference at the training parameter.

Both regimes train at $\alpha = \beta = 3.21$, the intermediate difficulty of (3.6), chosen in the nonlinear range where spectral bias should be most visible (WANG et al., 2024; ZHENG et al., 2023). Setting the two side by side isolates the effect of the physics term with the architecture and the budget held fixed.

3.4.3 Learning the Operator: FNO

The third rung changes the target of learning. Instead of one solution, the Fourier Neural Operator learns the map from parameter to solution for the entire family (3.2) in a single training, so a new $\alpha = \beta$ then costs one forward pass rather than a fresh optimization. Its architecture is the general operator of Section 2.4.2: a pointwise lifting takes the six input features at each grid point, the four channels of (3.4) together with the two normalized grid coordinates, up to a channel width of 32; four Fourier layers each combine a spectral convolution truncated to 16 modes per axis with a local 1×1 convolution and a GELU activation; and a pointwise projection returns the two output channels. This gives 2,106,146 trainable parameters, an order of magnitude above the pointwise models. Training minimizes the supervised mean-squared error (2.90) on the

normalized fields with Adam at learning rate 10^{-3} for 15,000 iterations; with only $M = 20$ samples, each iteration is effectively a full-batch step.

One implementation choice is worth pausing on, because it drives a result in Chapter 4. Each Fourier layer applies a two-dimensional real DFT over the last two axes of the feature array, which are the spatial *and the temporal* axes. The operator thus treats time as a second periodic spectral dimension and emits the whole space-time solution in one pass. That keeps the architecture uniform, but it revives the periodicity assumption of Section 2.4.2: the Boussinesq solution is not periodic in time, so the implicit wrap-around at $t = T$ excites the Gibbs phenomenon (GIBBS, 1899). We record the resulting boundary artifact as a measurable behaviour, and it later separates the FNO from its physics-informed successor.

3.4.4 Physics on the Operator: PINO

The top rung adds the equation to the operator, completing the ladder by combining the two previous additions. The Physics-Informed Neural Operator keeps the FNO backbone of Section 3.4.3 untouched and changes only the objective, appending the Boussinesq residual (2.96) to the data loss. Evaluating that residual means differentiating the predicted space-time array, and, following the reference implementation (LI et al., 2023b), we treat the two directions differently.

Spatial derivatives are taken pseudospectrally with the wavenumbers ξ_k of (2.18): we multiply the spatial DFT of the predicted field by $i\xi_k$ for the first derivative and by $-\xi_k^2$ for the second, then transform back. Temporal derivatives are taken by finite differences on the grid, second-order central in the interior and first-order one-sided at the two temporal boundaries. The mixed term u_{xxt} is formed by taking the spatial second derivative spectrally and then applying the temporal stencil. The residual is assembled in *physical* units: the normalized output and coefficients are mapped back through the inverse of (3.5) before the derivatives are formed, so the constraint enforces the true Boussinesq balance rather than a rescaled surrogate.

One further detail sets the residual apart from the data term. The supervised loss is evaluated on the native 128×128 training grid, but the physics residual is evaluated on a finer 256×256 grid, onto which the predicted field is spectrally interpolated before the derivatives are formed. A denser residual grid places a stricter, better-resolved constraint on the high-wavenumber content, at the price of extra transforms per iteration, and it means the physics term already sees resolution 256 during training, a fact we return to when reading the zero-shot resolution behaviour in Chapter 4. The objective is the weighted sum

$$\mathcal{L}_{\text{PINO}} = \lambda_{\text{pde}} \mathcal{L}_{\text{pde}} + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}}, \quad (3.12)$$

with the interior residual, the initial-condition term and the supervised term of (2.96). We weight the physics residual well above the other two, $\lambda_{\text{pde}} = 100$ against $\lambda_{\text{ic}} = 1$, so that the equation constraint leads the optimization; this emphasis was needed for the residual to leave a visible mark against the operator’s strong pull toward the data. As with the PINN, the data weight is the single toggle:

- **PINO (with data):** $\lambda_{\text{pde}} = 100$, $\lambda_{\text{ic}} = \lambda_{\text{data}} = 1$; physics and data act together, the residual dominating the objective and the data term anchoring the solution to the reference.
- **PINO (no data):** $\lambda_{\text{pde}} = 100$, $\lambda_{\text{ic}} = 1$, $\lambda_{\text{data}} = 0$; the operator is guided by the equation and the initial condition only, subject to the normalization caveat of Section 3.2.

Optimization is again Adam at learning rate 10^{-3} for 15,000 iterations, and every remaining hyperparameter matches the FNO, so any change in behaviour traces back to the physics term alone.

3.5 Probing the Spectral Bias

Each rung of the ladder shares one behaviour, the spectral bias derived in Section 2.3.4, and its cause deserves a direct examination rather than an inference from the training curves. We examine it with two probes of the same mechanism. The first looks at the Neural Tangent Kernel of a real Boussinesq PINN and asks whether the theory’s assumptions hold there. The second builds a controlled model in which those assumptions hold exactly, so that the theory’s quantitative predictions can be checked with nothing approximated. Together they connect the abstract eigenvalue analysis to the behaviour measured on the full architectures.

3.5.1 The Kernel of a Real PINN

The first probe inspects the Neural Tangent Kernel of the PINN residual directly. We train three networks of increasing width, $N_{\text{MLP}} \in \{8, 128, 512\}$ with four hidden layers, at $\alpha = \beta = 3.21$ on the physics-only objective, using stochastic gradient descent with a small learning rate (10^{-5}), the regime in which the frozen-kernel approximation (2.49) is expected to hold. The empirical kernel $K = JJ^\top$ is built explicitly at a fixed set of 40 probe points: automatic differentiation supplies the parameter gradient of each Boussinesq residual, and their Gram matrix (2.48) is the kernel. Every 500 iterations we log three quantities, the relative parameter drift $\|\theta - \theta_0\|/\|\theta_0\|$, the relative kernel drift $\|K - K_0\|/\|K_0\|$, and the eigenvalue spectrum of K at initialization and at the end of

training. The first two ask whether the kernel really stays put, as lazy training assumes; the third exposes the eigenvalue disparity that (2.63) names as the engine of spectral bias.

3.5.2 A Toy Model Where the Theory Is Exact

The kernel probe can only check whether the theory’s assumptions nearly hold, because a real network freezes its kernel only approximately. The second probe removes that gap. We build a small model in which the frozen-kernel dynamics of Section 2.3.4 hold exactly, so that the predicted error decay, spectral decay and per-mode iteration counts can be compared against measurement with nothing left to interpretation.

The construction is a Fourier-feature model, linear in its trainable parameters, fit to a periodic target on a regular grid. We sample a periodic target on N equispaced points and map each point through a fixed bank of Fourier features, sines and cosines at a range of wavenumbers; the only trainable parameters are the weights of a single linear layer on top of those features. Because the output is linear in those weights, the error Jacobian J does not depend on the parameters at all: it equals its initial value J_0 for the whole run, and the empirical kernel $K_0 = J_0 J_0^\top$ is genuinely constant rather than slowly drifting. Lazy training is then a property of the model by construction, not an approximation.

We train the model by full-batch gradient descent on the mean-squared error, so a single step is exactly

$$\theta \leftarrow \theta - \mu J_0^\top e, \quad (3.13)$$

the linearized update (2.60) with no higher-order correction. We set the learning rate to $\mu = 1/\lambda_{\max}$, with $\lambda_{\max}$ the largest eigenvalue of K_0 , which sits safely inside the stability bound $\mu < 2/\lambda_{\max}$ of Section 2.3.4, so every eigendirection contracts monotonically. Under these choices the error obeys $e^{(n)} = (I - \mu K_0)^n e_0$ of Equation (2.61) exactly, and the three predictions of the theory become testable one at a time:

- **Error decay** $e(\tau)$. We project the measured error onto each eigenvector v_k of K_0 and compare the amplitude against the predicted $(1 - \mu\lambda_k)^n$ of (2.61), following each eigendirection down to the float64 floor.
- **Spectral decay** $\hat{e}(\tau)$. We take the discrete Fourier transform of the error and compare the measured per-mode amplitude against the predicted decay of (2.69), checking that low wavenumbers resolve first and high wavenumbers last.
- **Per-mode iteration count** n_k . For each mode we record the iteration at which its amplitude first drops below a target precision ε and compare it against the closed-form estimate $n_k = \lceil \ln(|c_k(0)|/\varepsilon)/(\mu\lambda_k) \rceil$ of (2.73).

These predictions open the results of Chapter 4, where they anchor the reading of the spectral-bias curves measured on the full models.

3.6 Configuration Summary

Table 3.2 gathers the configuration of every experiment in one place. The four learned architectures were trained under a common budget, 15,000 iterations and, for the Adam-optimized models, learning rate 10^{-3} , so that the comparison of Chapter 4 reflects the architecture and the training regime rather than the effort spent optimizing.

Table 3.2 – Configuration of the five training experiments and the NTK diagnostic. The FNO and PINO share the same operator backbone and therefore the same parameter count; the MLP and PINN share the same feed-forward backbone and are single-parameter models. All Adam-trained models use 15,000 iterations.

Setting	MLP	PINN	FNO	PINO	NTK probe
Trainable parameters	198,401	198,658	2,106,146	2,106,146	8/128/512 width
Hidden layers	4	4	4 Fourier	4 Fourier	4
Width / channels	256	256	32	32	8, 128, 512
Fourier modes (per axis)	—	—	16	16	—
Optimizer	Adam	Adam	Adam	Adam	SGD
Learning rate	10^{-3}	10^{-3}	10^{-3}	10^{-3}	10^{-5}
Iterations	15,000	15,000	15,000	15,000	15,000
Parameters trained	single (3.21)	single (3.21)	all 20	all 20	single (3.21)
Collocation / batch	full batch	5,000 pts	full batch	full batch	40 probe pts
Loss weights $\lambda_{\text{pde}}/\lambda_{\text{ic}}/\lambda_{\text{data}}$	— / — / 1	1/1/{1, 0}	— / — / 1	100/1/{1, 0}	1/1/0
Physics-residual grid	—	5,000 pts	—	256×256	40 pts
Data regimes	data only	with / without	supervised	with / without	physics only

Source: The author (2026).

4 NUMERICAL RESULTS

This chapter reads the experiments built in Chapter 3 and lets them speak in a single arc. The arc begins with what each model costs to train (Section 4.1), then pins down the one behaviour they all share by carrying spectral bias from an exact toy calculation to the kernel of a real network (Section 4.2). With the mechanism established, the arc follows the error through the frequency bands as training proceeds (Section 4.3), fixes a common parameter to compare the final spectra fairly (Section 4.4), and isolates the one place where the physics term pays for itself (Section 4.5). Two stress tests then push the operators off their training conditions (Section 4.6), and Section 4.7 draws the threads together. The pseudospectral solver is the ground truth throughout, and every model leaves a failure mode that the arc makes explicit.

4.1 The Price of Each Approach

Before asking how well the models perform, we ask what they cost. Table 4.1 lists the size, wall-clock training time and final training loss of the six trained models, and two axes organize it. Along the first, capacity, the operators carry about 2.1 million parameters, roughly ten times the $\sim 200,000$ of the pointwise MLP and PINN. Along the second, and the more revealing one, training time, the models split sharply. The MLP finishes in under three minutes, and the plain FNO and the two PINNs in fifteen to twenty. The PINO stands apart at well over an hour, because its physics residual is assembled on the finer 256×256 grid (Section 3.4.4) at every iteration.

That hour buys generality, which is the return that justifies the operators' cost. A single FNO or PINO covers the entire 20-point family (3.2) from one training, whereas each MLP and PINN is welded to $\alpha = \beta = 3.21$ and would need retraining from scratch for any other value. Spread over the family, the operators, and the FNO above all, are the cheaper route to a solution at a new parameter, since inference there is a single forward pass.

One caution belongs up front, because the final-loss column invites a ranking it cannot support. The losses do not share a scale: each objective sums a different subset of data, residual and initial-condition terms. The PINO with data reaches a larger final loss

Table 4.1 – Model size, training time on the GTX 1660 SUPER, and final training loss. The final-loss column is *not* comparable across rows: each objective combines data, residual and initial-condition terms differently, so its magnitude reflects the makeup of the loss rather than solution accuracy. Accuracy is measured instead by the relative-error metrics of the following sections.

Model	Regime	Parameters	Train time (s)	Final loss
MLP	data only	198,401	142.8	3.5×10^{-5}
FNO	supervised	2,106,146	1103.8	3.6×10^{-6}
PINO	with data	2,106,146	5236.2	8.1×10^{-5}
PINO	no data	2,106,146	4221.6	4.7×10^{-5}
PINN	with data	198,658	884.4	7.9×10^{-5}
PINN	no data	198,658	861.3	1.2×10^{-4}

Source: The author (2026).

(8.1×10^{-5}) than the PINO without (4.7×10^{-5}) because the extra supervised term adds its own residual to the total, and the sections below show it is in fact the more accurate of the two. The plain FNO carries the smallest loss in the table (3.6×10^{-6}) because its objective is a single data term with nothing added. Training loss is a poor proxy for accuracy here, and every comparison that follows rests on the held-out relative-error metrics of Section 3.3.

4.2 Spectral Bias, From Theory to the Network

The models share one behaviour above all others, and before comparing them we establish its mechanism. Spectral bias, derived in Section 2.3.4, predicts that gradient training resolves low frequencies before high ones at a rate set by the eigenvalues of the Neural Tangent Kernel. We confirm this in two steps. First we check the prediction where it can be made exact, on the toy Fourier-feature model of Section 3.5.2. Then we look for the same mechanism in the kernel of a real Boussinesq PINN, where the assumptions hold only approximately.

The toy model removes every approximation, so it is the right place to test the theory quantitatively. Its Jacobian is constant by construction, so the frozen-kernel dynamics hold exactly and the closed-form predictions of Section 2.3.4 can be compared against measurement with nothing left to interpretation. Figure 4.1 reports the three checks.

Each panel confirms one prediction. In Figure 4.1a the amplitude of the error along each eigendirection follows the predicted $(1 - \mu\lambda_k)^n$ of (2.61): the measured markers sit on the theoretical curves down to the float64 floor, and the directions with large eigenvalues collapse first while those with small eigenvalues linger, the per-eigendirection ordering

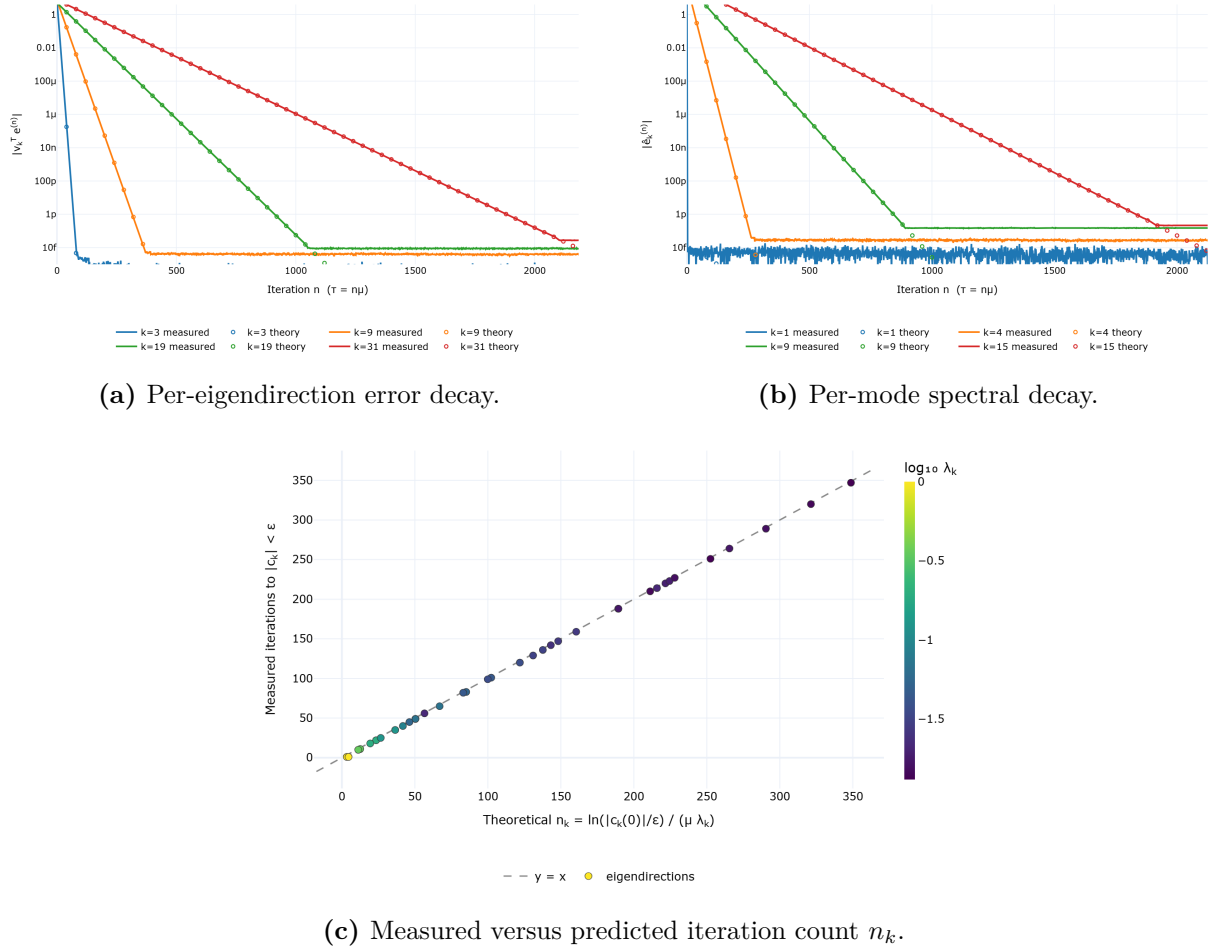


Figure 4.1 – Spectral-bias theory validated on the Fourier-feature model of Section 3.5.2, trained by full-batch gradient descent at $\mu = 1/\lambda_{\max}$ so that the linearized update (2.61) is exact. Measured curves (markers) overlay the theoretical predictions (lines) until the float64 floor.

Source: The author (2026).

of (2.59). Figure 4.1b reads the same dynamics in Fourier space, where the measured per-mode amplitude tracks the prediction of (2.69) and the low wavenumbers resolve first, the frequency principle stated literally. Figure 4.1c makes the agreement quantitative: plotting, for each mode, the measured iteration at which its amplitude first falls below the target precision against the closed-form estimate n_k of (2.73) places the points on the $y = x$ diagonal, coloured by $\log_{10} \lambda_k$, with the smallest-eigenvalue modes demanding the most iterations. The theory holds to numerical precision on the model built to satisfy its assumptions.

With the prediction confirmed in the exact case, we look for the same mechanism in a real network, whose kernel drifts. Figure 4.2 shows the diagnostic of Section 3.5.1 on the Boussinesq PINN residual for three network widths.

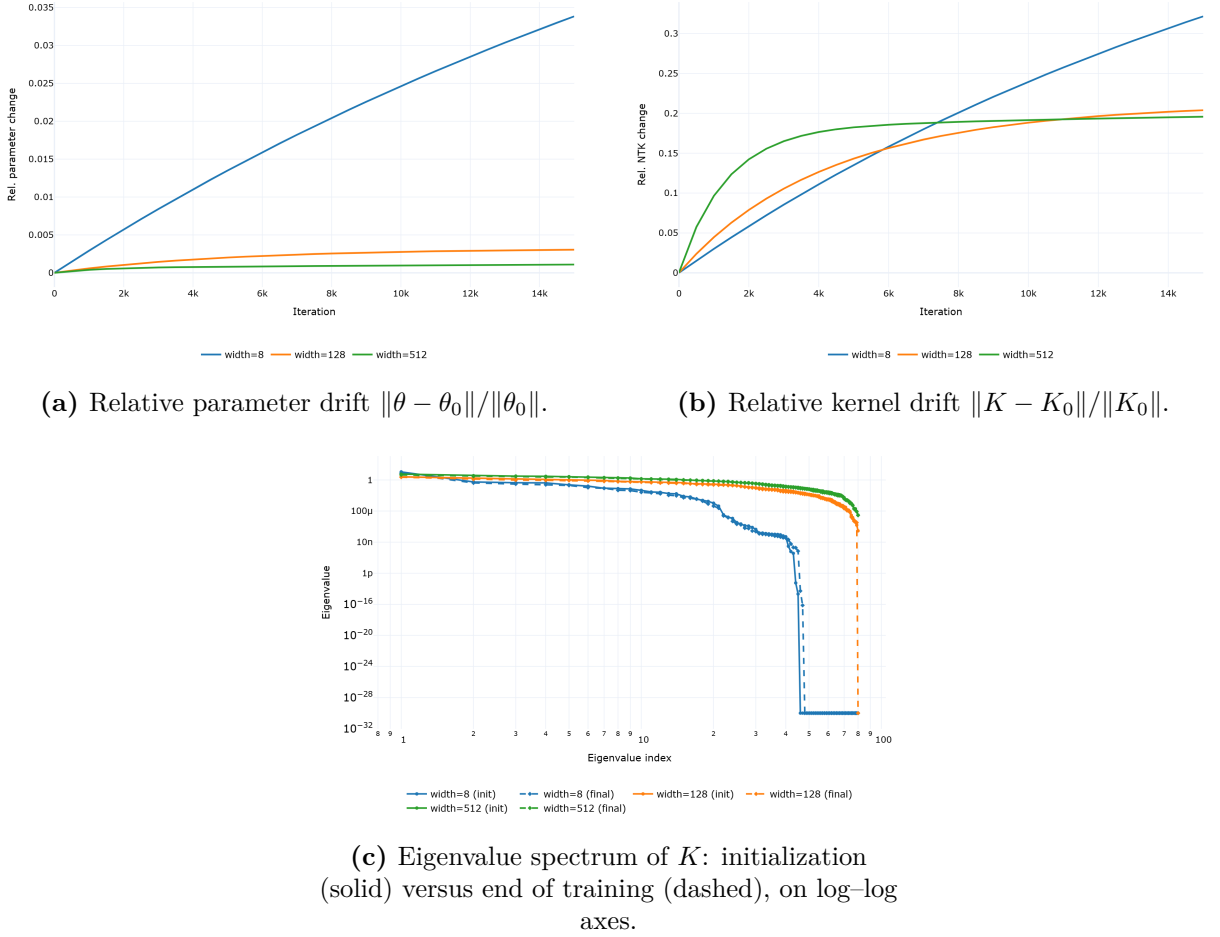


Figure 4.2 – Neural Tangent Kernel diagnostic for the Boussinesq PINN residual at $\alpha = \beta = 3.21$, for widths 8, 128 and 512.

Source: The author (2026).

The diagnostic confirms both ingredients the theory depends on. The frozen-kernel assumption behind (2.49) sharpens as the network widens: the width-512 network drifts

by only about 0.06% in its parameters and about 12% in its kernel over the whole run, while the width-8 network drifts by several percent in parameters and far more in its kernel, and the curves order themselves monotonically in width. The eigenvalue spectrum of the last panel then supplies the driver: it spans many orders of magnitude and falls off steeply, with the width-8 kernel collapsing to the numerical floor after roughly 45 modes and the wider kernels decaying more gradually but still steeply. By (2.63) the time to resolve an eigendirection is inversely proportional to its eigenvalue, so this steep decay means every direction past the leading smooth ones is learned prohibitively slowly. This is the eigenvalue disparity that Wang, Yu, et al. (2022) name as the source of spectral bias, and it sets the expectation for everything that follows: the network should pick up the low-frequency structure readily and stall on the high-frequency detail. The exact toy confirmed the prediction; the real kernel shows the same mechanism at work, and the band-tracking curves of the next section can be read as that mechanism operating inside the full architectures.

4.3 Learning by Frequency Band

We now watch spectral bias develop as the optimizer runs. Figure 4.3 tracks the relative spectral error in the low, mid and high bands of (3.11) across training for all six models. Within any single panel the figure tells one clean story; across panels it carries the caveat of Section 3.3, that the operators are monitored on the near-linear sample $\alpha = \beta = 0.1$ while the pointwise MLP and PINN are monitored on the harder $\alpha = \beta = 3.21$. Cross-family magnitudes are therefore not directly comparable in this figure, and the apples-to-apples comparison waits for Section 4.4, where every model is evaluated at the common parameter 3.21.

The within-panel ordering is the same in all six cases and admits no exception: the low band is driven lowest, the mid band trails, and the high band is learned last and least. Every architecture and every regime obeys this ordering, which is the measured face of the NTK spectrum of Figure 4.2 and the toy validation of Figure 4.1. What varies from panel to panel is the size of the gap and how far each band is pushed down.

The MLP panel (Figure 4.3a) is the purest specimen, which is why we read it first. With no physics term and no initial-condition term competing against the data fit, its ordering is spectral bias with no confounders. The low band falls from 0.837 at the first logged epoch to 0.015 at the last, almost two orders of magnitude, while the mid band falls from 1.004 to 0.114 and the high band barely stirs, moving only from 1.009 to 0.704. The low frequencies are crushed and the high frequencies are left almost untouched, which is the prediction of Section 2.3.4 in its cleanest empirical form. Every other panel is a variation on it with data or physics added.

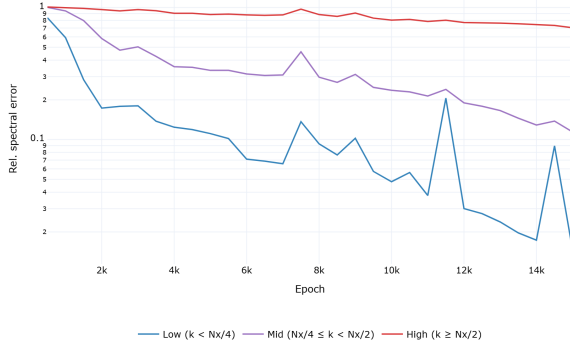
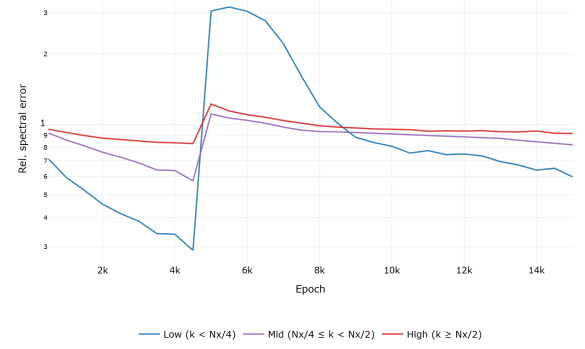
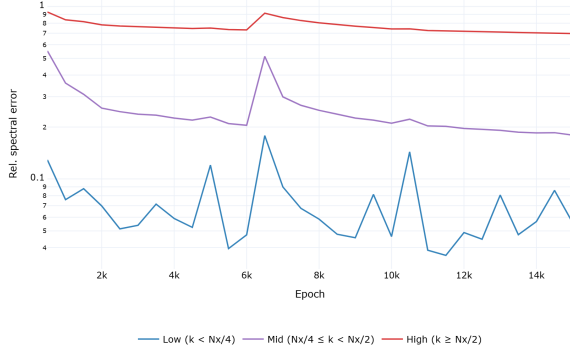
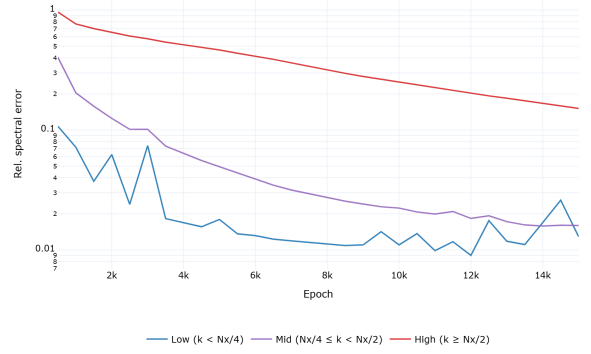
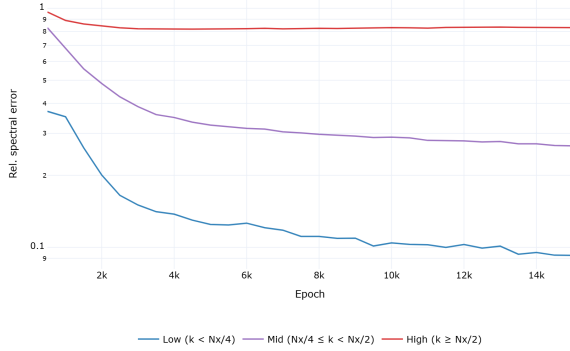
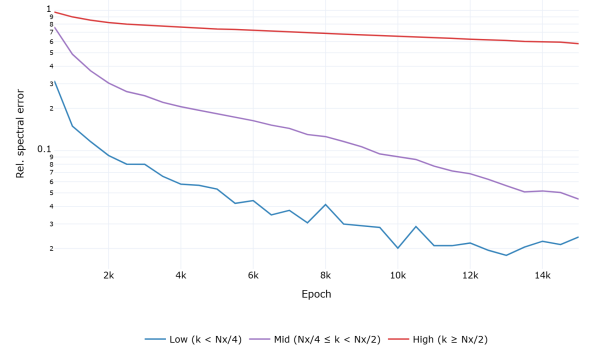
(a) MLP (data only), at $\alpha = \beta = 3.21$.(b) PINN (no data), at $\alpha = \beta = 3.21$.(c) PINN (with data), at $\alpha = \beta = 3.21$.(d) FNO, at $\alpha = \beta = 0.1$.(e) PINO (no data), at $\alpha = \beta = 0.1$.(f) PINO (with data), at $\alpha = \beta = 0.1$.

Figure 4.3 – Relative spectral error by frequency band during training. In every panel the low band (blue) is learned first and driven lowest, the mid band (purple) trails it, and the high band (red) is learned last and least, the signature of spectral bias. Note the mixed reference samples: the pointwise MLP and PINN are tracked at $\alpha = \beta = 3.21$, the operators at $\alpha = \beta = 0.1$.

Source: The author (2026).

The comparisons that *are* valid across panels hold the architecture and the reference sample fixed and change only the data term, and they are decisive. For the PINN at 3.21, adding data lowers the terminal low-band error from about 0.13 to about 0.06 and the mid band from about 0.55 to about 0.18, while leaving the high band near 0.70 almost unchanged. For the PINO at 0.1 the effect is larger: the data term drives the low band from about 0.37 to about 0.02 and the mid band from about 0.83 to about 0.05, again with a much smaller gain in the high band, which falls only from about 0.83 to about 0.58. In both families the supervised term is a strong lever on the low and mid bands and a weak one on the high band. The physics residual alone, in the two no-data panels, never brings any band near the levels reached with data, and the PINO (no data) high band scarcely moves over the whole run, staying above 0.83.

The FNO panel, monitored at the same easy parameter as the PINO, reaches the lowest low- and mid-band errors in the figure (about 0.013 and 0.016) and the lowest high-band error too (about 0.15). Even so, its high band sits an order of magnitude above its low band. Read across its panels, Figure 4.3 says that operators do not *solve* spectral bias, they *compress* it: the best model still leaves a high-frequency residual of roughly 15%, and every other model leaves more.

4.4 Reproducing the Spectrum

To compare the families on equal terms we fix $\alpha = \beta = 3.21$ and inspect the full spatial spectrum of the prediction at three times, $t = 0$, $t = T/2 = 15$ and $t = T = 30$. Figures 4.4–4.9 report the six models; in every panel the black curve is the reference spatial spectrum and the dashed orange curve is the prediction, with the relative-error curves and the time-averaged error at the foot of each figure.

The pointwise panels lay the pathology bare, and they carry it into the initial state. Both PINN regimes already fail to represent the high-wavenumber content of the sech^2 profile at $t = 0$, and the relative error climbs toward unity as the mode index grows. Without data (Figure 4.5) the predicted spectrum has shed most of its mid- and high-frequency amplitude by $t = 30$, and the mean spectral error reaches 0.70. Adding data (Figure 4.6) tightens the low-wavenumber crests, in step with the low-band gain of Figure 4.3, yet the aggregate mean barely moves ($0.17 \rightarrow 0.58 \rightarrow 0.68$), because the mid and high modes dominate it and the data term fails to recover them at this hard parameter. The MLP (Figure 4.4) tells the same story with the physics stripped away: it fits the reference where the frequencies are low and undershoots where they are high, its mean spectral error rising from about 0.16 at $t = 0$ to about 0.51 at $t = 15$ and about 0.65 at $t = 30$. Data helps the pointwise models where the kernel is favourable and does not rescue them where it is adverse.

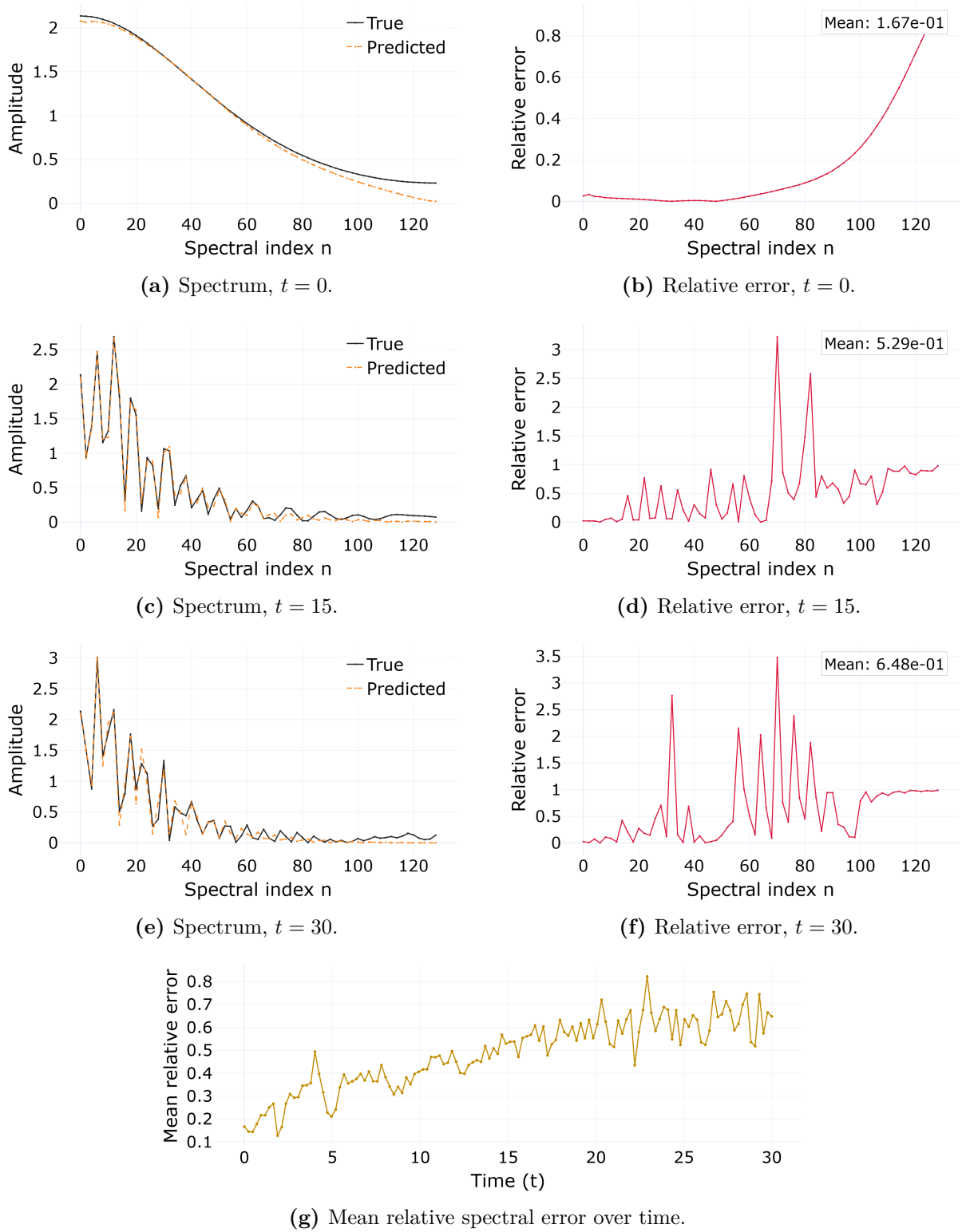


Figure 4.4 – MLP (data only) spatial spectrum at $\alpha = \beta = 3.21$. Fitting the reference with no equation and no physics term, the network reproduces the low-wavenumber crests but undershoots the tail: the relative error climbs with wavenumber, the purest picture of spectral bias among the six models.

Source: The author (2026).

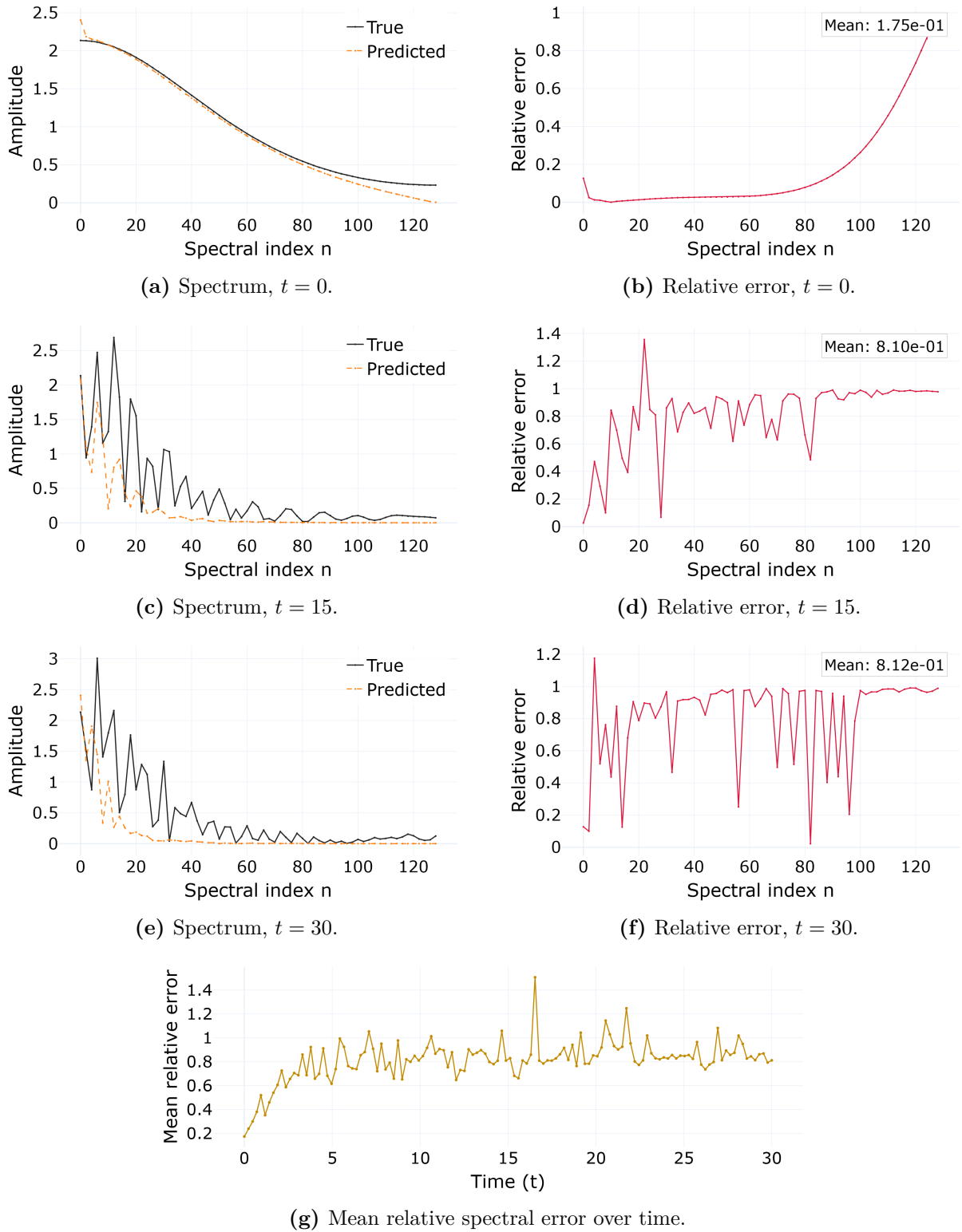
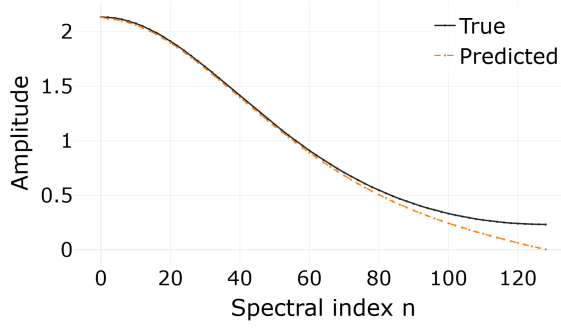
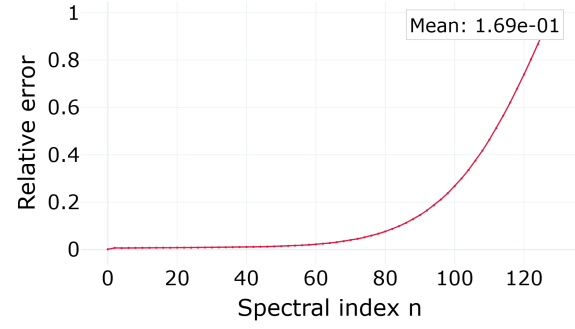
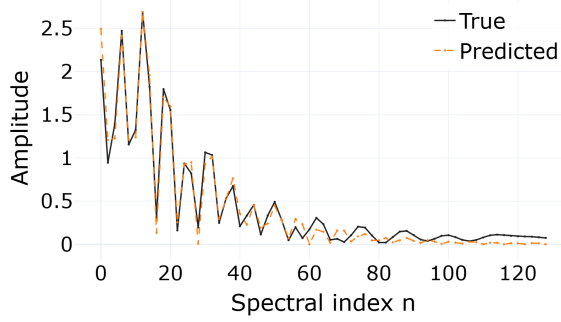
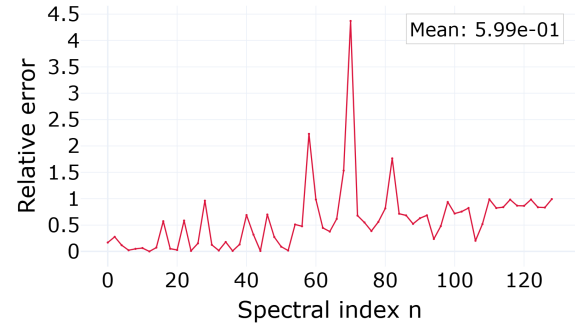
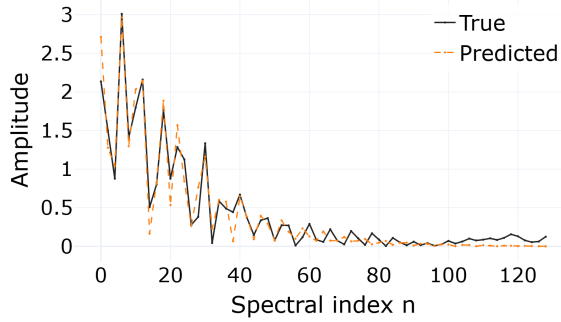
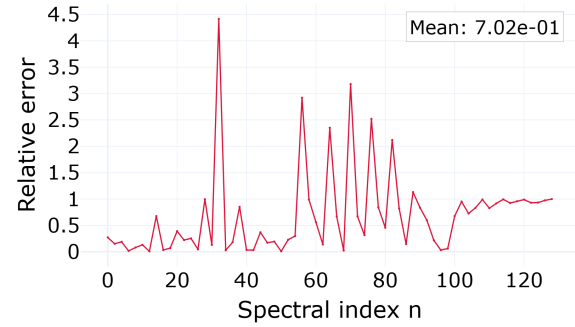
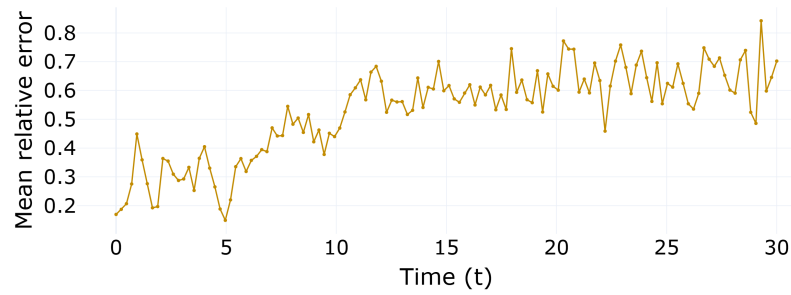


Figure 4.5 – PINN (no data) spatial spectrum at $\alpha = \beta = 3.21$. The network cannot represent the sech^2 tail even at $t = 0$, and the relative error rises monotonically with wavenumber; the mean spectral error grows from 0.17 at $t = 0$ to 0.70 at $t = 30$.

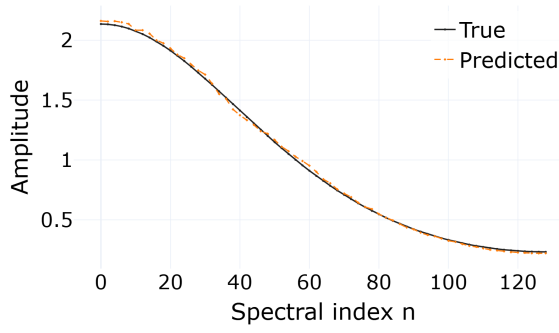
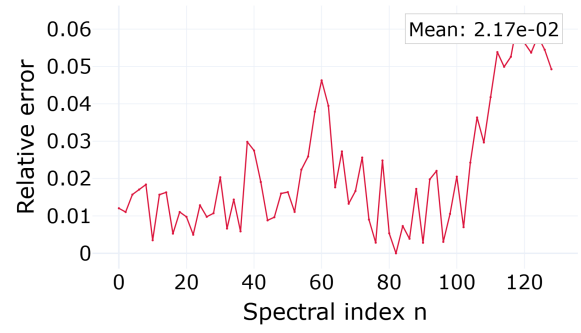
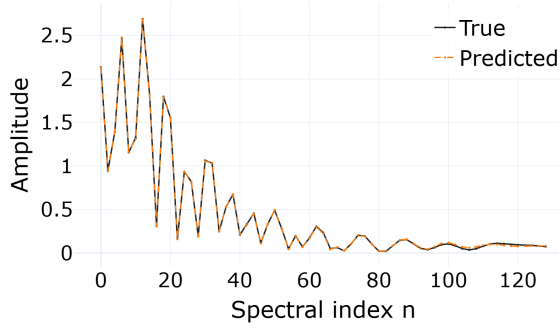
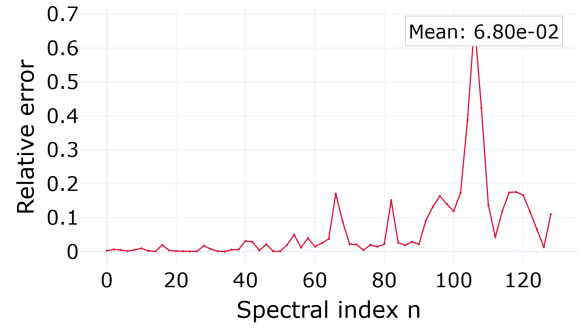
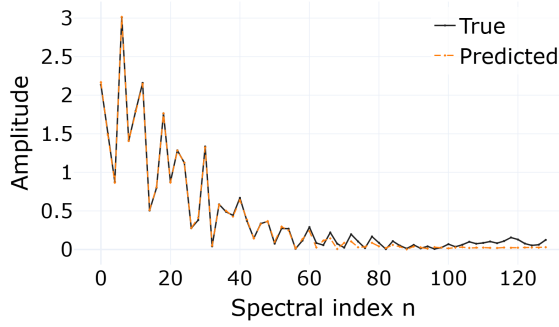
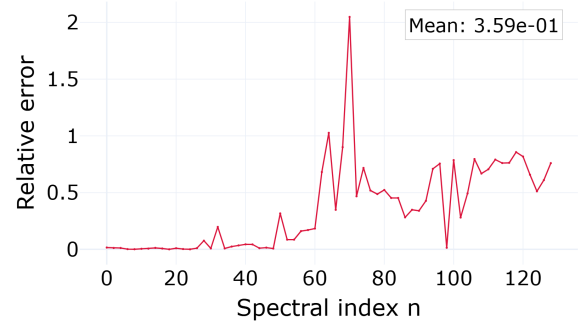
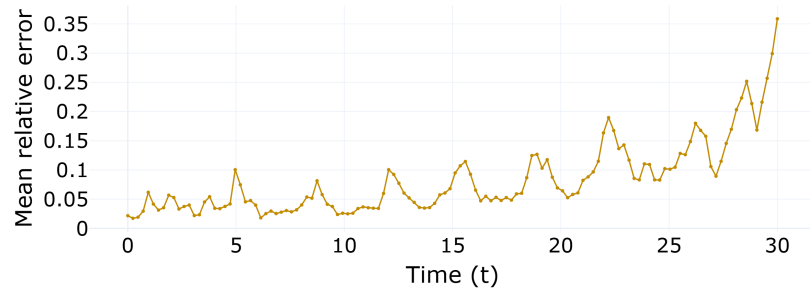
Source: The author (2026).

(a) Spectrum, $t = 0$.(b) Relative error, $t = 0$.(c) Spectrum, $t = 15$.(d) Relative error, $t = 15$.(e) Spectrum, $t = 30$.(f) Relative error, $t = 30$.

(g) Mean relative spectral error over time.

Figure 4.6 – PINN (with data) spatial spectrum at $\alpha = \beta = 3.21$. The low-wavenumber crests are tracked more tightly than in Figure 4.5, but the high-wavenumber undershoot and the growth of the mean error in time ($0.17 \rightarrow 0.58 \rightarrow 0.68$) remain.

Source: The author (2026).

(a) Spectrum, $t = 0$.(b) Relative error, $t = 0$.(c) Spectrum, $t = 15$.(d) Relative error, $t = 15$.(e) Spectrum, $t = 30$.(f) Relative error, $t = 30$.

(g) Mean relative spectral error over time.

Figure 4.7 – FNO spatial spectrum at $\alpha = \beta = 3.21$. Low and mid wavenumbers are tracked almost exactly (mean error 0.013 at $t = 0$), but the error migrates to the high modes and grows steadily in time, reaching a mean of 0.40 at $t = 30$.

Source: The author (2026).

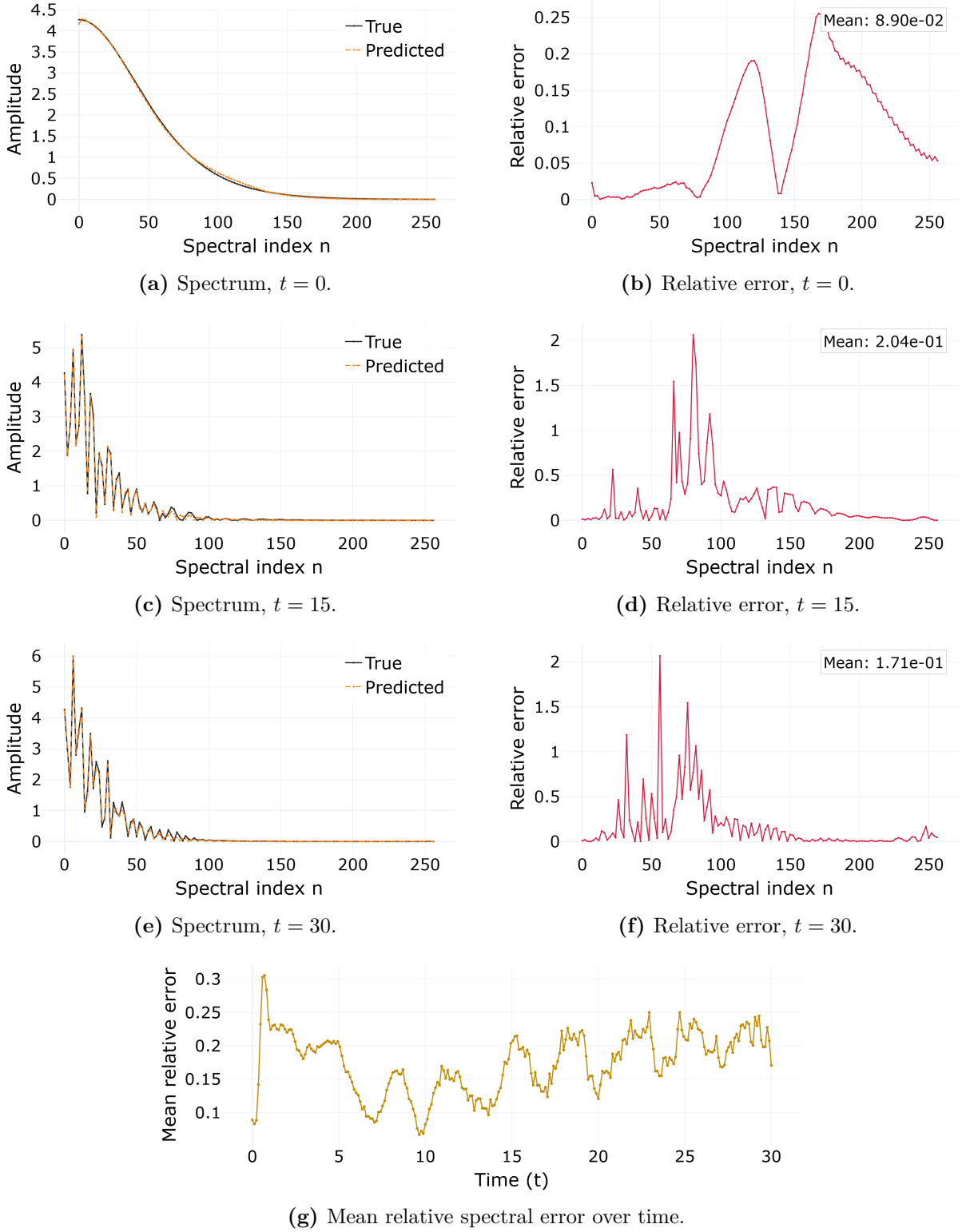


Figure 4.8 – PINO (with data) spatial spectrum at $\alpha = \beta = 3.21$. The prediction tracks the reference across the band and, unlike the FNO, the mean error stays bounded in time ($0.077 \rightarrow 0.13 \rightarrow 0.20$): the physics residual steadies the temporal evolution.

Source: The author (2026).

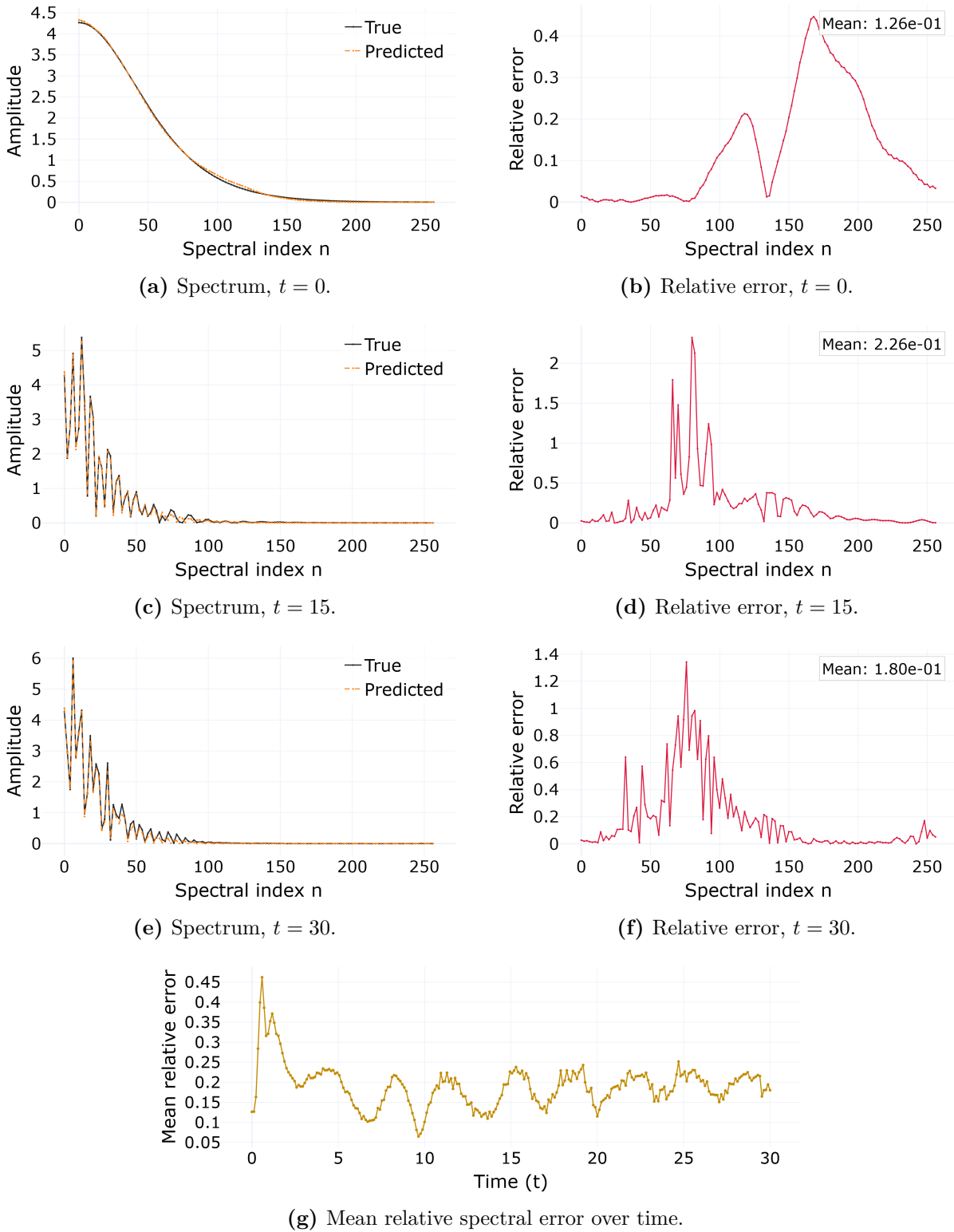


Figure 4.9 – PINO (no data) spatial spectrum at $\alpha = \beta = 3.21$. Without the supervised term the low- and mid-band undershoot returns (mean error $0.11 \rightarrow 0.39 \rightarrow 0.33$), confirming that the physics residual alone does not overcome spectral bias.

Source: The author (2026).

The operators, learning in the frequency domain, invert this picture at low and mid wavenumbers. The FNO (Figure 4.7) reproduces the low and mid spectrum almost exactly at early times, a mean error of 0.013 at $t = 0$, an order of magnitude below the pointwise models, because those modes are precisely what its spectral layers parametrize. Its weakness lies in time: the error gathers in the high modes and *grows*, from a mean of 0.013 at $t = 0$ to 0.40 at $t = 30$. The PINO with data (Figure 4.8) inherits the operator’s spectral accuracy without this temporal decay, its mean error rising only from 0.077 to 0.20 over the same horizon, the steadiest of the six at late times. The contrast is sharp at $t = 30$, where the PINO with data (0.20) clearly beats the FNO (0.40) though the FNO was ahead at $t = 0$: the physics residual trades a slightly worse start for a much steadier trajectory. The PINO without data (Figure 4.9) forfeits the advantage, its low- and mid-band undershoot returning and its mean error climbing to 0.33–0.39, midway between the operators-with-data and the PINNs. The physics constraint by itself does not replace data.

4.5 Where Physics Earns Its Keep

The temporal growth of the FNO error in Figure 4.7 has a specific and instructive cause, and isolating it shows the one place where the physics term pays for itself. As Section 3.4.3 explains, the operator applies the Fourier transform along the temporal axis as well as the spatial one, implicitly assuming the trajectory is periodic in time. It is not: the field at $t = T$ does not match the field at $t = 0$, so the assumption manufactures an artificial discontinuity at the temporal boundary and excites the Gibbs phenomenon there (GIBBS, 1899). Figures 4.10 and 4.11 make the effect and its cure visible.

In the FNO panel (Figure 4.10) every parameter shows the same feature: after an initial transient the time-resolved error settles to a plateau and then jumps at $t = 30$ to two or three times that plateau. The spike appears at all three nonlinearities and is the dominant part of the FNO’s late-time error. In the PINO panel (Figure 4.11) the predicted surfaces are visually identical, yet the spike is gone and the error rises only slightly toward the boundary. The PDE residual and the initial-condition term penalize exactly the temporal inconsistency that a periodic transform would otherwise tolerate, damping the boundary oscillation. This is the one place in the study where the physics term delivers a clear, concrete benefit: it suppresses the FNO’s Gibbs artifact. The benefit is a suppression and not an elimination, since a small terminal rise remains, and it arrives at the modelling cost of the residual machinery.

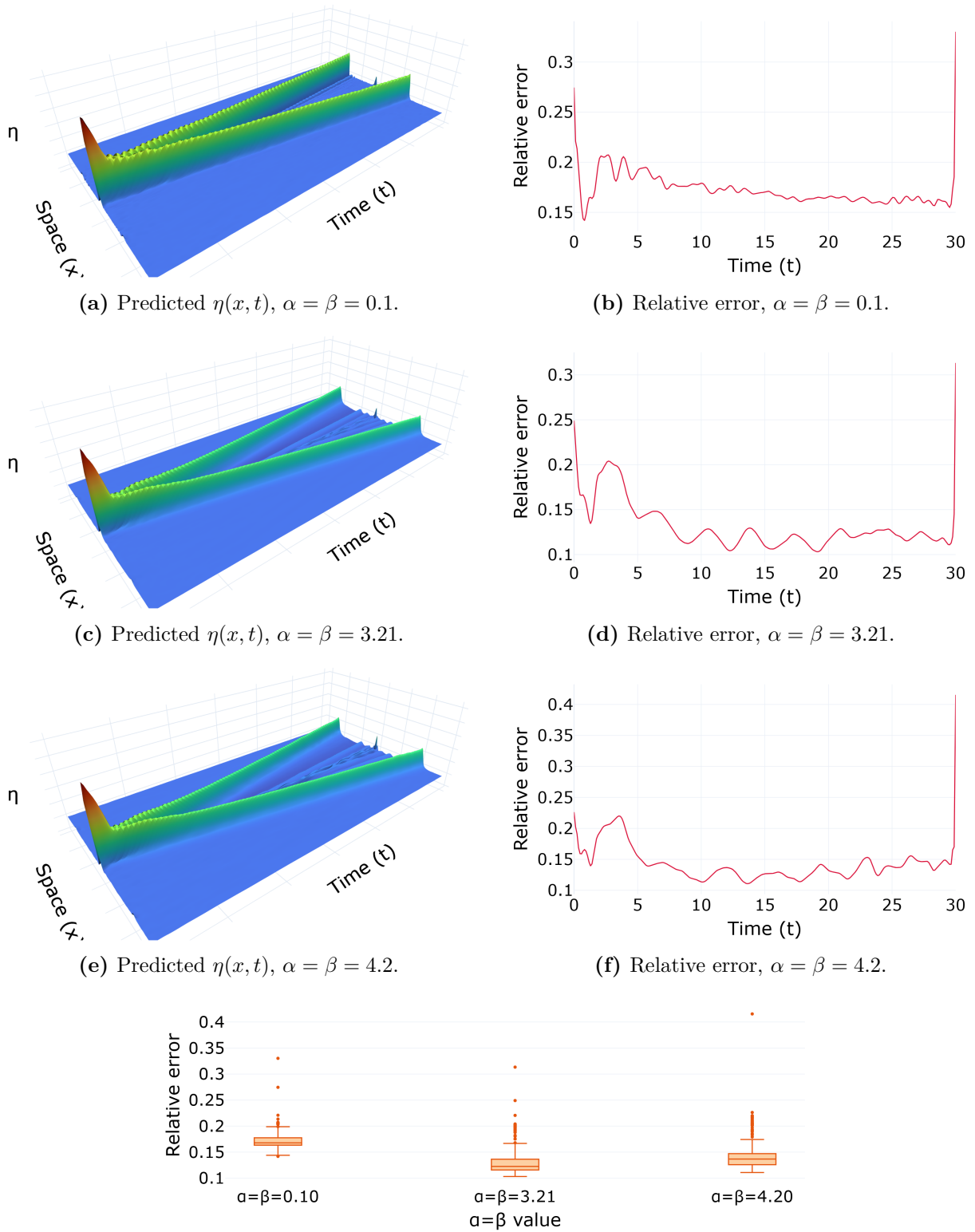


Figure 4.10 – FNO across the nonlinearity axis ($\alpha = \beta \in \{0.1, 3.21, 4.2\}$), evaluated at resolution 256. The predicted surfaces (left) capture the two counter-propagating pulses, but the time-resolved relative error (right) spikes sharply at the final time $t = 30$: the temporal Gibbs artifact of treating time as a periodic axis.

Source: The author (2026).

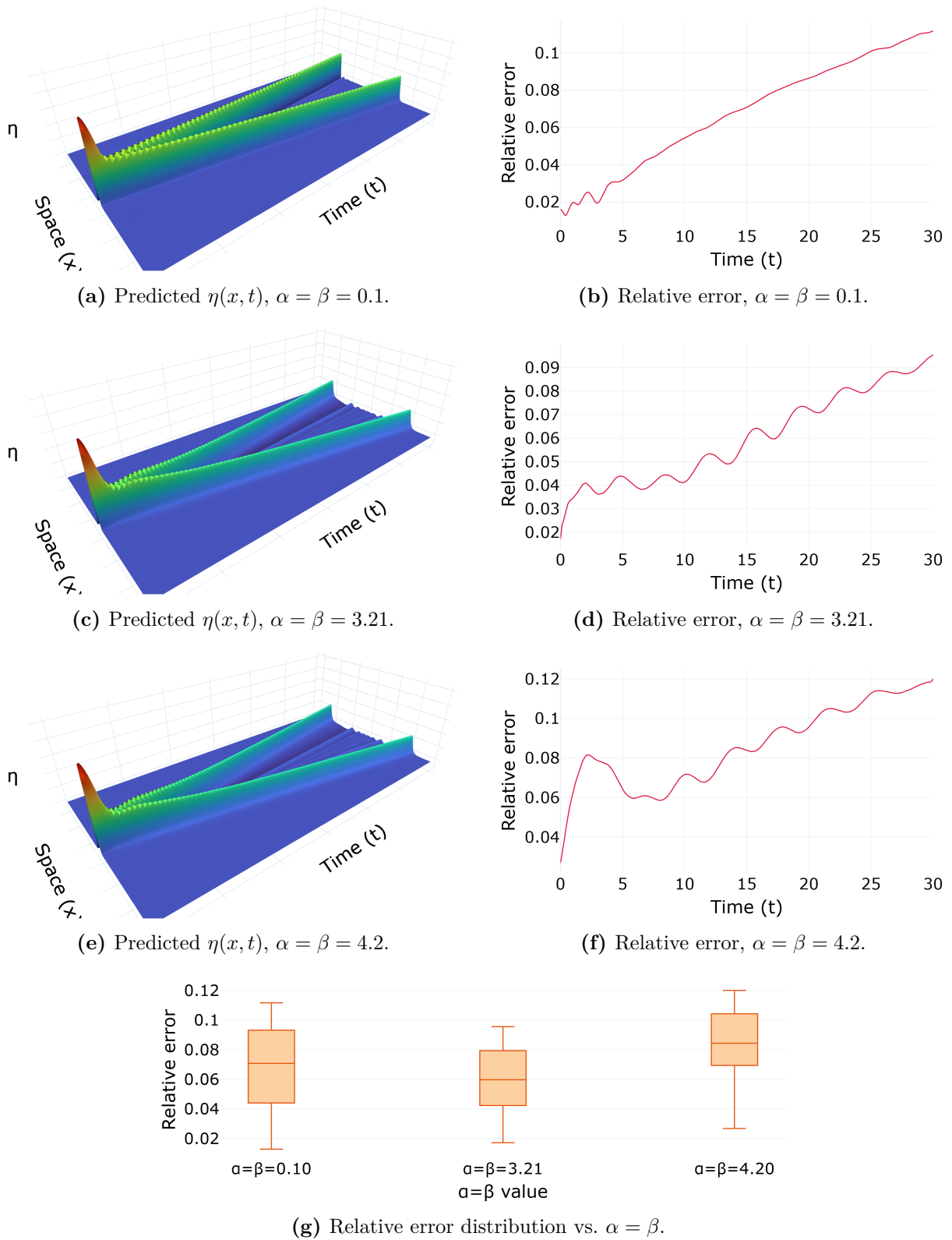


Figure 4.11 – PINO (with data) across the same nonlinearity axis and resolution. The predicted surfaces are visually indistinguishable from the FNO, but the terminal-time spike of Figure 4.10 is gone: the error rises only gently toward $t = 30$. The physics residual and the initial-condition term act as a temporal regularizer.

Source: The author (2026).

4.6 Two Stress Tests

The comparisons so far held the operators on their training conditions. Two stress tests now push them off those conditions, one along the nonlinearity axis and one along the query resolution. Both concern the operators alone, since each pointwise MLP and PINN is trained at a single parameter and a single resolution and has nothing to say about either sweep.

The first test sweeps the nonlinearity. The box plots at the foot of Figures 4.10 and 4.11 summarize the distribution of the time-resolved error (3.8) over the whole trajectory at each of the three parameters. Their most striking feature runs against intuition: the *smallest* nonlinearity, $\alpha = \beta = 0.1$, carries the *largest* relative error for both operators, a median near 0.18 against roughly 0.12–0.13 at the more nonlinear values 3.21 and 4.2. We read this carefully. Two effects plausibly combine. First, $\alpha = \beta = 0.1$ sits at the extreme lower edge of the training grid (3.2), where the operator has the least surrounding data to interpolate from, so a boundary-of-range penalty is expected. Second, the relative metric (3.8) divides by the instantaneous norm of the reference: in the near-linear regime the solution is dominated by the two clean, slowly deforming pulses of the wave-equation limit, whose energy is concentrated and whose norm is modest, so a fixed absolute error reads as a larger relative one. The nonlinear cases spread energy across the domain and are, in relative terms, more forgiving. The box plots do not show the operators growing uniformly more accurate, in absolute terms, as nonlinearity rises. They show only that the relative error, under this normalization and this training grid, is non-monotone in the parameter, with the near-linear boundary case the least well served.

The second test queries the operator off its training resolution, probing the discretization-invariance operators are meant to enjoy. Figure 4.12 takes the PINO with data, trained at resolution 128, and evaluates it at 128, 256 and 512 at the fixed parameter $\alpha = \beta = 3.21$ with no retraining.

The geometric promise holds. At every resolution the operator returns a smooth, coherent surface with the correct pulse structure, with no retraining and no architectural change; the input is simply resampled on the finer grid. The accuracy promise is subtler. The box plot shows the relative error *increasing* with resolution, from a median near 0.07 at the training resolution of 128 to about 0.12 at 256 and 0.16 at 512. The cause is that the finer references contain genuine high-wavenumber content that the operator, trained at 128 and keeping only 16 Fourier modes per axis, never learned to produce, so querying it against a sharper reference exposes the fine scales it is missing. A secondary effect appears at the training resolution itself, where the error oscillates in time in step with the wave reflections, the aliasing signature of a coarse grid. Zero-shot superresolution is real as a *geometric* property, since the operator accepts and returns fields at any resolution, and

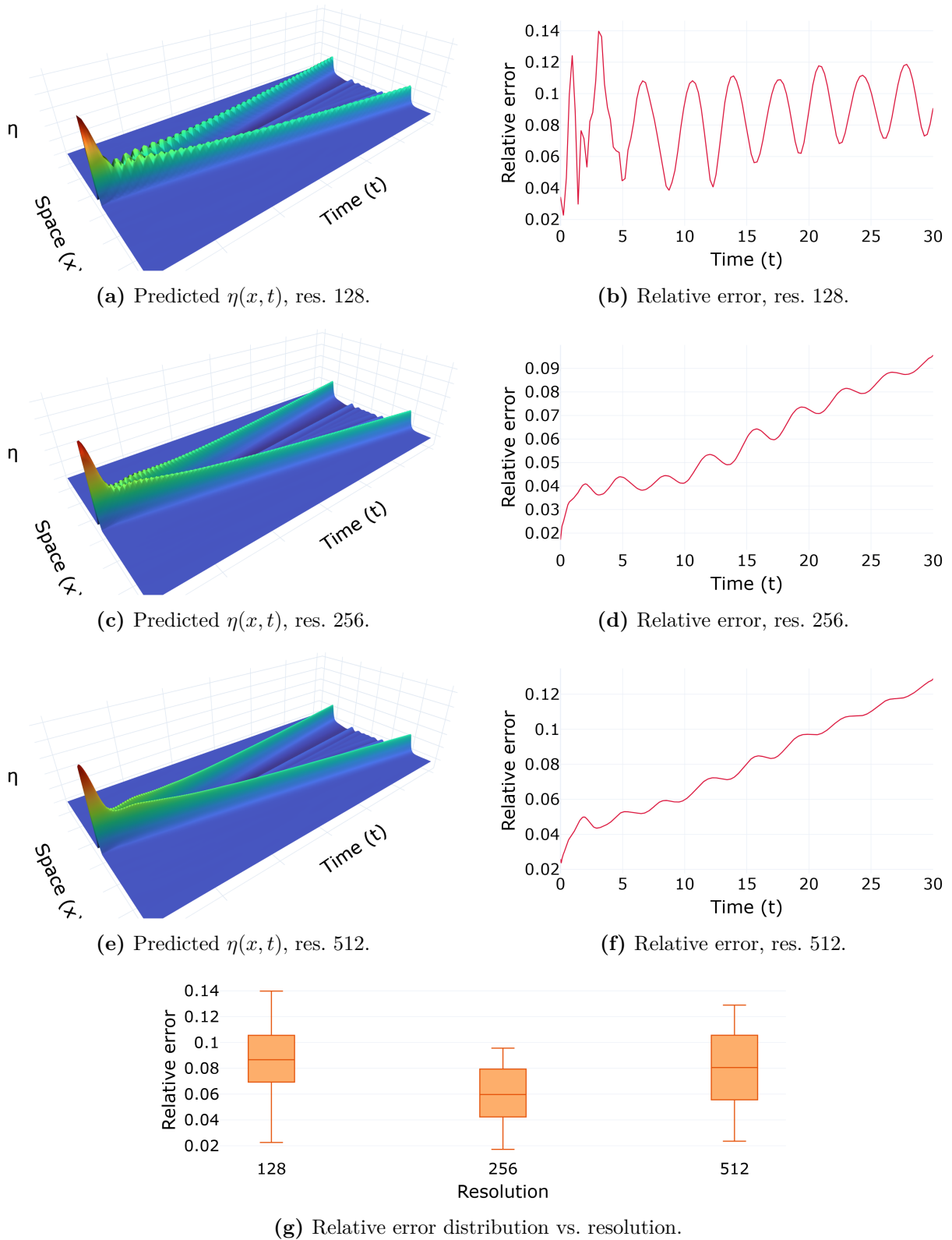


Figure 4.12 – PINO (with data) queried zero-shot at resolutions 128, 256 and 512 at $\alpha = \beta = 3.21$. The operator returns a coherent field at every resolution, but the relative error against the pseudospectral reference *grows* with resolution (median rising from about 0.07 at 128 to about 0.16 at 512): invariance of form is not invariance of accuracy.

Source: The author (2026).

it is not an *accuracy* property: fidelity is anchored to the spectral band seen in training, and a richer grid can only reveal, never recover, the modes left out.

4.7 What the Experiments Show

Taken together, the experiments do not crown a single winner; they chart where each method succeeds and where it fails.

- **Spectral bias is universal, and provably so here.** The toy validation of Section 4.2 matches the predicted per-mode decay and iteration counts to numerical precision, the real PINN kernel shows the same eigenvalue disparity, and every architecture then obeys the same ordering: low wavenumbers first and best, high wavenumbers last and least (Figures 4.2, 4.1, 4.3). The data-only MLP is the purest instance, driving its low band by nearly two orders of magnitude while its high band barely moves. No method drives the high band below roughly 0.15 relative error, and the best there is the plain FNO.
- **Data is the decisive lever, and physics is not a substitute for it.** Within both the PINN and the PINO families, adding the supervised term sharply cuts the low- and mid-band error, while the physics-only regimes revert toward the biased baseline (Figures 4.3, 4.8, 4.9). Physics alone neither matches data nor removes the bias.
- **Operators dominate at low and mid wavenumbers.** At the common parameter 3.21 the FNO and the PINO with data track the low and mid spectrum an order of magnitude more accurately than either PINN or the MLP (Section 4.4), and they cover the whole family from one training, whereas each pointwise model is bound to a single parameter.
- **Physics buys temporal stability.** The FNO’s error grows in time and spikes at the temporal boundary through the Gibbs artifact of its periodic time axis, and the PINO’s residual suppresses the spike and keeps the late-time error bounded (Section 4.5). This is the clearest concrete benefit of the physics term in this study.
- **The remaining limitations are real.** The near-linear boundary parameter is the least accurate (Section 4.6), zero-shot superresolution degrades in accuracy as the query grid sharpens (Section 4.6), and the high-frequency band is never fully resolved by any method. These are consistent features of the design, not artefacts of a single run.

The picture matches the one anticipated in Chapter 1: these methods complement the pseudospectral solver that sets the accuracy ceiling here rather than replacing it. Their value is in amortizing the cost of the parameter sweep and in the temporal regularization that physics supplies, and it is least of all at the high wavenumbers where the reference solver stays unchallenged.

5 CONCLUSION

This work built, from first principles, a pseudospectral Boussinesq solver and three families of Scientific Machine Learning methods, Physics-Informed Neural Networks, Fourier Neural Operators and Physics-Informed Neural Operators, and set them against one another on a single controlled testbed: the one-dimensional Boussinesq system with a fixed solitary initial condition and a coupled coefficient $\alpha = \beta$ swept from the near-linear to the strongly nonlinear regime, with the pseudospectral solution as both training reference and accuracy ceiling. One finding organizes all the others. Spectral bias is universal across every architecture and regime we tried, supervised data is the strongest lever against it, the physics residual buys temporal stability rather than raw accuracy, and the finest scales remain the property of the classical solver alone. The sections below turn this single finding into a verdict on each family, state what the study cannot claim, and point to the work that should come next.

5.1 A Verdict on Each Approach

The models divide into two families by how a single trained network relates to the parameter axis, and reading them family by family also answers the four questions posed in Chapter 1.

The pointwise family: MLP and PINN.

These networks represent one solution as a continuous map and are welded to a single parameter, $\alpha = \beta = 3.21$. They are where spectral bias shows its cleanest face. The data-only MLP, with no equation and no initial-condition term competing against the data fit, drove its low band down by nearly two orders of magnitude while its high band barely moved, spectral bias with no confounders left in the objective, which answers the third research question in its starkest form: the bias is present, and here it is isolated. Adding the equation, in the PINN, did not repair the high band; both PINN regimes failed to represent the tail of the sech^2 spectrum even at $t = 0$ and undershot the mid and high modes throughout. Within this family the data term was the decisive influence, tightening the low-wavenumber crests, while the physics residual on its own

reverted toward the biased baseline, which is the pointwise half of the answer to the second research question. Because these models were run only at $\alpha = \beta = 3.21$, their behaviour as nonlinearity grows is left open; what we can say is bounded to that one regime. Against the pseudospectral baseline of the first research question, the pointwise family never closed the gap in the mid and high bands, and it covers only the parameter it was trained on.

The operator family: FNO and PINO.

These operators learn the whole family (3.2) from a single training and answer a new parameter with one forward pass. They invert the pointwise picture at low and mid wavenumbers, tracking that part of the spectrum an order of magnitude more accurately than either PINN or the MLP, and this is the operators' side of the first research question: closer to the baseline where the kernel is favourable, and covering the family that the pointwise models cannot. The second research question resolves most sharply here. Data remained the strongest lever, since the no-data PINO forfeited the low- and mid-band accuracy that the with-data PINO reached, yet the physics residual earned its keep in a specific place, suppressing the temporal-boundary Gibbs spike that degrades the plain FNO at the final time and holding the late-time error bounded. The most effective configuration was the hybrid, data and physics together, combining the operator's spectral accuracy with the residual's temporal regularization. The operators still exhibit spectral bias at the highest wavenumbers, compressing it further than the pointwise models without removing it, which completes the third research question: the bias survives every regime, and no method drove the high band below roughly 15% relative error. One further operator finding qualifies the promise of discretization-invariance. Querying the trained PINO at finer grids returned a coherent field at every resolution, so the invariance holds as a geometric property. The relative error still grew with resolution, because a richer grid cannot recover the fine scales absent from training, so it does not hold as an accuracy property.

The mechanism, common to both families.

The fourth research question asked whether Neural Tangent Kernel theory genuinely predicts the frequency imbalance or only redescribes it, and the two probes of Chapter 4 answer that it predicts. On the Boussinesq PINN the kernel drifted less as the network widened, approaching the frozen-kernel regime the analysis assumes, and its eigenvalue spectrum decayed across many orders of magnitude, the disparity that forces the high-frequency directions to be learned prohibitively slowly. On a Fourier-feature

model whose kernel is constant by construction, the measured per-eigendirection decay, per-mode spectral decay and closed-form iteration counts matched the theory to numerical precision. This ties the frequency-band errors of every architecture to a concrete mechanism rather than a loose analogy, and it lets us read the compressed-but-persistent high-band error, in both families, as the same eigenvalue disparity at work.

5.2 What This Study Cannot Claim

Several boundaries fence in these conclusions and should be weighed before carrying them further.

- **One initial condition and a coupled parameter.** Every experiment used a single solitary profile $\eta(x, 0) = \text{sech}^2(x)$ and held $\alpha = \beta$. The operators therefore learned a one-parameter family, not the full four-parameter Boussinesq family, and the reported generalization is generalization along that single coordinate. Decoupling α and β and varying the initial data would test a substantially harder mapping.
- **Pointwise models seen at one parameter.** The MLP and the PINN were each trained and evaluated only at $\alpha = \beta = 3.21$. Their difficulty in the mid and high bands is a statement about that regime, and it is not evidence of how they would behave as nonlinearity grows, which would require training them across the axis.
- **Time as a spectral axis.** The operator implementation applies the Fourier transform along the temporal axis, treating time as periodic, which is the direct cause of the terminal-time Gibbs artifact of Section 4.5. A time-marching or autoregressive operator, as suggested by Li et al. (2020), would drop the periodicity assumption entirely.
- **Normalization coupling in the physics-only regime.** As noted in Section 3.2, the output normalization constants come from the reference dataset, so the no-data models are not strictly data-independent: they inherit the output scale of the reference. A fully data-free study would fix these constants from the initial condition alone.
- **Fixed capacity and budget.** Every operator kept only 16 Fourier modes per axis and trained for 15,000 iterations at a single learning rate. The persistent high-band error is consistent with this truncation, and a mode or capacity sweep would be needed to separate the architectural limit from the optimization limit.

- **Relative-error normalization.** The non-monotone dependence of the error on nonlinearity (Section 4.6) is partly an artefact of normalizing by the instantaneous solution norm. Absolute or energy-based metrics would round out the account of accuracy across the parameter range.
- **Single-parameter pointwise models.** Because each MLP and PINN encodes one solution, the pointwise family contributes nothing to the nonlinearity and resolution sweeps, so those findings rest on the operators alone.

5.3 Directions Forward

The results point to several concrete continuations. The most immediate is to attack the shared failure mode directly: since spectral bias survives every regime and its mechanism is now pinned to the kernel eigenvalue disparity, the levers that target that disparity are the ones to test on this same benchmark, namely adaptive loss balancing (WANG; TENG, et al., 2021), Fourier feature embeddings, and the second-order and spectrum-aware optimizers of Khodakarami et al. (2026). A second direction is architectural: a time-marching FNO would remove the temporal Gibbs artifact by construction, isolating how much of the PINO’s late-time advantage is intrinsic to the physics term rather than a repair of a periodicity assumption. A third is to broaden the problem by decoupling α and β , varying the initial data, and extending to the two-dimensional Boussinesq system, to see whether the ordering of methods found here survives a genuinely higher-dimensional solution operator. A fourth is to widen the field of comparison to DeepONet (LU et al., 2021) and to hybrid schemes that hand the high-frequency residual back to a classical corrector, which would probe whether the complementary role identified for these methods can become a solver that is at once fast and spectrally faithful.

For the nonlinear Boussinesq system, the value of these Scientific Machine Learning methods lies in amortizing a parameter sweep and, when physics is added, in gaining temporal stability, while the fine scales that make dispersive waves hard remain, for now, the solver’s alone.

REFERENCES

ARORA, S. et al. Fine-Grained Analysis of Optimization and Generalization for Overparameterized Two-Layer Neural Networks. In: PROCEEDINGS of the 36th International Conference on Machine Learning (ICML). [S.l.: s.n.], 2019. v. 97, p. 322–332.

BAKER, N. et al. **Workshop Report on Basic Research Needs for Scientific Machine Learning: Core Technologies for Artificial Intelligence**. Washington, DC, 2019. DOI: 10.2172/1478744.

BIHLO, A.; POPOVYCH, R. O. Physics-informed neural networks for the shallow-water equations on the sphere. **Journal of Computational Physics**, 2022. Available from: <<https://arxiv.org/abs/2104.00615>>.

BONA, J. L.; CHEN, M.; SAUT, J.-C. Boussinesq equations and other systems for small-amplitude long waves in nonlinear dispersive media: I. Derivation and linear theory. **Journal of Nonlinear Science**, v. 12, p. 283–318, 2002. DOI: 10.1007/s00332-002-0466-4.

BONEV, B. et al. Spherical Fourier Neural Operators: Learning Stable Dynamics on the Sphere. In: INTERNATIONAL Conference on Machine Learning (ICML). [S.l.: s.n.], 2023. Available from: <<https://arxiv.org/abs/2306.03838>>.

BORDELON, B.; CANATAR, A.; PEHLEVAN, C. Spectrum Dependent Learning Curves in Kernel Regression and Wide Neural Networks. In: PROCEEDINGS of the 37th International Conference on Machine Learning (ICML). [S.l.: s.n.], 2020. Available from: <<https://arxiv.org/abs/2002.02561>>.

BOUSSINESQ, J. Théorie des ondes et des remous qui se propagent le long d'un canal rectangulaire horizontal. **Journal de Mathématiques Pures et Appliquées**, v. 17, p. 55–108, 1872.

BRUNTON, S. L.; PROCTOR, J. L.; KUTZ, J. N. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. **Proceedings of the National Academy of Sciences**, v. 113, n. 15, p. 3932–3937, 2016. DOI: 10.1073/pnas.1517384113.

BURDEN, R. L.; FAIRES, J. D. **Numerical Analysis**. 9. ed. Boston: Brooks/Cole, 2011.

CAO, Y. et al. Towards Understanding the Spectral Bias of Deep Learning. In: PROCEEDINGS of the Thirtieth International Joint Conference on Artificial Intelligence (IJCAI). [S.l.: s.n.], 2021. Available from: <<https://arxiv.org/abs/1912.01198>>.

CHEN, R. T. Q. et al. Neural Ordinary Differential Equations. In: ADVANCES in Neural Information Processing Systems (NeurIPS). [S.l.: s.n.], 2018. v. 31. Available from: <<https://arxiv.org/abs/1806.07366>>.

CHEN, T.; CHEN, H. Universal approximation to nonlinear operators by neural networks with arbitrary activation functions and its application to dynamical systems. **IEEE Transactions on Neural Networks**, v. 6, n. 4, p. 911–917, 1995.

CUOMO, S. et al. Physics-informed neural networks for data-driven simulation: Advantages, limitations, and opportunities. **Physica A: Statistical Mechanics and its Applications**, 2023. DOI: 10.1016/j.physa.2022.128415.

CYBENKO, G. Approximation by superpositions of a sigmoidal function. **Mathematics of Control, Signals and Systems**, v. 2, n. 4, p. 303–314, 1989.

DE PINA, I. et al. Investigating Steady Unconfined Groundwater Flow using Physics Informed Neural Networks. **Advances in Water Resources**, 2023. DOI: 10.1016/j.adwatres.2023.104445.

_____. Investigating Steady Unconfined Groundwater Flow Using Physics-Informed Neural Networks. **arXiv preprint arXiv:2112.13792**, 2021. Available from: <<https://arxiv.org/abs/2112.13792>>.

DEBNATH, L. **Nonlinear Partial Differential Equations for Scientists and Engineers**. 3. ed. New York: Birkhäuser, 2012.

ENGSIK-KARUP, A. P. et al. Fourier pseudospectral methods for 2D Boussinesq-type equations. **Ocean Modelling**, 2012. DOI: 10.1016/j.ocemod.2012.05.003.

GIBBS, J. Willard. Fourier's Series. **Nature**, v. 59, n. 1539, p. 606, 1899. DOI: 10.1038/059606a0.

GOLUB, G. H.; VAN LOAN, C. F. **Matrix Computations**. 4. ed. Baltimore: Johns Hopkins University Press, 2013.

GUPTA, G. et al. Multiwavelet-based Operator Learning for Differential Equations. In: **ADVANCES in Neural Information Processing Systems (NeurIPS)**. [S.l.: s.n.], 2021. Available from: <<https://arxiv.org/abs/2109.13459>>.

HORNIK, K.; STINCHCOMBE, M.; WHITE, H. Multilayer feedforward networks are universal approximators. **Neural Networks**, v. 2, n. 5, p. 359–366, 1989.

JACOT, A.; GABRIEL, F.; HONGLER, C. Neural Tangent Kernel: Convergence and Generalization in Neural Networks. In: **ADVANCES in Neural Information Processing Systems (NeurIPS)**. [S.l.: s.n.], 2018. v. 31. Available from: <<https://arxiv.org/abs/1806.07572>>.

JAGTAP, A. D. et al. Deep Learning of Inverse Water Waves Problems Using Multi-Fidelity Data: Application to Serre–Green–Naghdi Equations. **arXiv preprint arXiv:2202.02899**, 2022. Available from: <<https://arxiv.org/abs/2202.02899>>.

KARNIADAKIS, G. E. et al. Physics-informed machine learning. **Nature Reviews Physics**, v. 3, n. 6, p. 422–440, 2021. DOI: 10.1038/s42254-021-00314-5.

KHODAKARAMI, S. et al. Spectral bias in physics-informed and operator learning: analysis and mitigation guidelines. **arXiv preprint**, 2026. Available from: <<https://arxiv.org/abs/2602.19265>>.

KINGMA, D. P.; BA, J. Adam: A method for stochastic optimization. **arXiv preprint arXiv:1412.6980**, 2014.

KREYSZIG, E. **Introductory functional analysis with applications**. [S.l.]: John Wiley & Sons, 1978.

KRISHNAPRIYAN, A. S. et al. Characterizing Possible Failure Modes in Physics-Informed Neural Networks. In: **ADVANCES in Neural Information Processing Systems (NeurIPS)**. [S.l.: s.n.], 2021. v. 34, p. 26548–26560.

KUMAR, S.; DHAMA, S. Physics-informed neural networks for exploring soliton collisions in the non-integrable Schamel equation in plasmas. **Physics of Fluids**, v. 37, n. 1, 2025. DOI: 10.1063/5.0301334.

LECUN, Y.; BENGIO, Y.; HINTON, G. Deep learning. **Nature**, v. 521, n. 7553, p. 436–444, 2015. DOI: 10.1038/nature14539.

LI, Z. et al. Fourier Neural Operator for parametric partial differential equations. In: **INTERNATIONAL Conference on Learning Representations (ICLR)**. [S.l.: s.n.], 2021. Available from: <<https://arxiv.org/abs/2010.08895>>.

_____. Physics-Informed Neural Operator for Learning Partial Differential Equations. **ACM/IMS Transactions on Data Science**, 2023. Available from: <<https://arxiv.org/abs/2111.03794>>.

_____. **PINO: Physics-Informed Neural Operator – Reference Implementation**. [S.l.: s.n.], 2023. GitHub repository. Available from: <<https://github.com/neuraloperator/PINO>>.

LI, Z. et al. Fourier neural operator for parametric partial differential equations. **arXiv preprint arXiv:2010.08895**, 2020.

LINNAINMAA, A. Taylor expansion of the accumulated rounding error. **BIT Numerical Mathematics**, v. 10, n. 1, p. 47–55, 1970.

LKHAGVASUREN, B.; CHOI, B.-J.; JIN, H. S. Applications of the Fourier neural operator in a regional ocean modeling and prediction. **Frontiers in Marine Science**, v. 11, 2024. DOI: 10.3389/fmars.2024.1383997.

LU, L. et al. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. **Nature Machine Intelligence**, v. 3, p. 218–229, 2021. DOI: 10.1038/s42256-021-00302-5.

MARTINS, L. G. **Estudo Numérico do Modelo Boussinesq com Topografia e Pressão Variável no Espaço e no Tempo**. 2023. MA thesis – Universidade Federal do Paraná, Curitiba.

MARTINS, L. G.; FLAMARION, M. V.; RIBEIRO-JR, R. A forced Boussinesq model with a sponge layer. **Partial Differential Equations in Applied Mathematics**, v. 10, p. 100661, 2024. DOI: 10.1016/j.padiff.2024.100661.

MCCULLOCH, W. S.; PITTS, W. A logical calculus of the ideas immanent in nervous activity. **Bulletin of Mathematical Biophysics**, v. 5, n. 4, p. 115–133, 1943.

NOCEDAL, J. Updating quasi-Newton matrices with limited storage. **Mathematics of Computation**, v. 35, n. 151, p. 773–782, 1980.

PARIKH, N.; BOYD, S. Proximal Algorithms. **Foundations and Trends in Optimization**, v. 1, n. 3, p. 123–231, 2013. DOI: 10.1561/24000000003.

PASZKE, A. et al. PyTorch: An Imperative Style, High-Performance Deep Learning Library. In: ANNUAL Conference on Neural Information Processing Systems (NeurIPS). Vancouver: [s.n.], 2019. p. 8024–8035. Available from: <<https://arxiv.org/abs/1912.01703>>.

PATEL, A. et al. A Neural Operator-Based Emulator for Regional Shallow Water Dynamics. **arXiv preprint**, 2025. Available from: <<https://arxiv.org/abs/2502.14782>>.

PEI, K. et al. Data-driven soliton mappings for integrable fractional nonlinear wave equations via deep learning with Fourier neural operator. **Chaos, Solitons & Fractals**, v. 165, 2022. DOI: 10.1016/j.chaos.2022.112848.

PEREGRINE, D. H. Long waves on a beach. **Journal of Fluid Mechanics**, v. 27, n. 4, p. 815–827, 1967. DOI: 10.1017/S0022112067002605.

RACKAUCKAS, C. et al. Universal Differential Equations for Scientific Machine Learning. **arXiv preprint arXiv:2001.04385**, 2020.

RAHAMAN, N. et al. On the spectral bias of neural networks. In: PMLR. INTERNATIONAL Conference on Machine Learning (ICML). Long Beach: [s.n.], 2019. p. 5301–5310.

RAISSI, M.; PERDIKARIS, P.; KARNIADAKIS, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. **Journal of Computational Physics**, v. 378, p. 686–707, 2019.

ROSENBLATT, F. The perceptron: a probabilistic model for information storage and organization in the brain. **Psychological Review**, v. 65, n. 6, p. 386–408, 1958.

ROSOFSKY, S. et al. Applications of Physics-Informed Neural Operators (PINOs): Shallow Water Equations and Beyond. **arXiv preprint**, 2023. Available from: https://github.com/shawnrosofsky/PINO_Applications.

RUMELHART, D. E.; HINTON, G. E.; WILLIAMS, R. J. Learning representations by back-propagating errors. **Nature**, v. 323, n. 6088, p. 533–536, 1986.

STRANG, G. **Linear Algebra and Its Applications**. 4. ed. Belmont: Thomson Brooks/Cole, 2006.

STROGATZ, S. H. **Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering**. 2. ed. Boca Raton: CRC Press, 2018.

SÜLI, E.; MAYERS, D. F. **An Introduction to Numerical Analysis**. Cambridge: Cambridge University Press, 2003.

TODO. A Review of Physics-Informed Machine Learning in Fluid Mechanics. **Energies**, v. 16, n. 5, 2023. DOI: 10.3390/en16052343.

TODO. Modelo Numérico de Boussinesq Adaptado para Ondas de Embarcação. **Revista Brasileira de Recursos Hídricos**, v. 15, n. 4, p. 5–15, 2010. DOI: 10.21168/rbrh.v15n4.p5-15.

TREFETHEN, L. N. **Spectral Methods in MATLAB**. Philadelphia: Society for Industrial and Applied Mathematics, 2000. DOI: 10.1137/1.9780898719598.

UCHIDA, T. et al. Ocean Emulation With Fourier Neural Operators: Double Gyre. **Journal of Advances in Modeling Earth Systems**, v. 17, n. 1, 2025. DOI: 10.1029/2023MS004137.

WANG, L. et al. Explorations of nonlinear waves of the Boussinesq and Camassa-Holm equations using PINNs. **Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences**, v. 480, n. 2286, 2024. DOI: 10.1098/rspa.2023.0580.

WANG, S.; TENG, Y.; PERDIKARIS, P. Understanding and Mitigating Gradient Flow Pathologies in Physics-Informed Neural Networks. **SIAM Journal on Scientific Computing**, v. 43, n. 5, a3055–a3081, 2021. DOI: 10.1137/20M1318043.

WANG, S.; YU, X.; PERDIKARIS, P. When and Why PINNs Fail to Train: A Neural Tangent Kernel Perspective. **Journal of Computational Physics**, v. 449, p. 110768, 2022. DOI: 10.1016/j.jcp.2021.110768.

WHITHAM, G. B. **Linear and Nonlinear Waves**. New York: Wiley-Interscience, 1974.

XU, Z.-Q. J. et al. Frequency Principle: Fourier Analysis Sheds Light on Deep Neural Networks. In: 5. COMMUNICATIONS in Computational Physics. [S.l.: s.n.], 2020. v. 28, p. 1746–1767. DOI: 10.4208/cicp.0A-2020-0085.

YU, J. et al. Spherical multigrid neural operator for improving autoregressive global weather forecasting. **Scientific Reports**, v. 15, n. 1, 2025. DOI: 10.1038/s41598-025-96208-y.

ZHANG, Y. et al. Influence of layer and neuron configurations on Boussinesq equation solutions via a bilinear neural network. **Nonlinear Dynamics**, v. 112, p. 12315–12333, 2024. DOI: 10.1007/s11071-024-09708-3.

ZHENG, S. et al. Prediction of phase transition and time-varying dynamics of the (2+1)-dimensional Boussinesq equation by parameter-integrated PINNs with phase domain decomposition. **Physical Review E**, v. 108, n. 4, 2023. DOI: 10.1103/PhysRevE.108.045303.

ZHONG, M.; YAN, Z.; TIAN, S. Data-driven parametric soliton-rogon state transitions for nonlinear wave equations using deep learning with Fourier neural operator. **Communications in Theoretical Physics**, v. 75, n. 2, p. 025001, 2023. DOI: 10.1088/1572-9494/acab55.